

Causal Feature Learning via Generalized Rayleigh Quotients

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Abstract

Extracting causally meaningful features from time-series data is fundamental for robust machine learning under distribution shifts. In process monitoring, existing methods struggle to maintain detection performance when operating conditions change. Current approaches capture either temporal causal relationships or cross-environment invariance, but not both simultaneously. We propose Causal Feature Learning (CFL), a unified framework that jointly optimizes for temporal relevance and environment mean invariance. CFL formulates feature extraction as a generalized Rayleigh-quotient problem, maximizing correlation with target variables while penalizing sensitivity to environment-dependent mean shifts. Theoretical analysis establishes conditions under which CFL identifies a mean-invariant predictive subspace. Experiments on the Tennessee Eastman Process demonstrate that CFL achieves 94.6% average fault detection rate, outperforming 15 baseline methods while operating at the lowest realized false-alarm rate and detection delay.

1. Introduction

Discovering causal relationships from observational time-series data is a fundamental challenge in machine learning and process engineering (Pearl, 2009; Spirtes et al., 2000). In industrial process monitoring, understanding the causal structure among process variables enables robust fault detection that generalizes across different operating conditions (Chiang et al., 2000; Qin, 2012). However, most existing methods focus on statistical correlations rather than causal relationships, leading to features that may fail when

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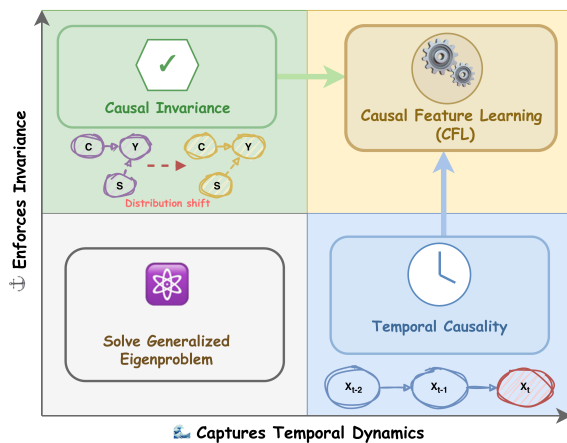


Figure 1. Two complementary aspects of causal features. (a) Temporal causality: time-lagged covariances encode how past variables help predict current states. (b) Causal invariance: causal mechanisms are expected to be stable across environments. CFL extracts features capturing both properties.

the underlying data distribution shifts.

Causal features in time-series data manifest through two complementary mechanisms, as illustrated in Figure 1. First, temporal causality reflects how past values of one variable influence current values of another, formalized through concepts like Granger causality (Granger, 1969). This temporal information is encoded in time-lagged covariance structures. If X_{t-k} helps predict and potentially causes Y_t , their lagged cross-covariance can be informative. Second, causal invariance reflects the principle that causal mechanisms remain autonomous across environments or operating conditions (Peters et al., 2017; Schölkopf et al., 2012). In this work, we operationalize invariance via first-order invariance across environments: spurious correlations often manifest as environment-dependent mean shifts, while causal mechanisms are expected to be more stable.

Existing dynamic latent variable methods capture these aspects separately. Slow Feature Analysis (SFA) (Wiskott & Sejnowski, 2002) extracts features that vary slowly over time, implicitly assuming that meaningful signals are temporally smooth while noise fluctuates rapidly. Dynamic methods like DiPCA (Dong & Qin, 2018a) and

DiCCA (Dong & Qin, 2018b) model temporal dependencies through autoregressive structures, capturing Granger-type predictive effects. However, neither approach explicitly optimizes for both temporal relevance and robustness to cross-environment mean shifts.

In this paper, we propose causal feature learning, a spectral framework that jointly captures temporal relevance and mean invariance through generalized Rayleigh-quotient optimization. Our formulation balances two objectives: maximizing predictive covariance with target variables one step ahead, and minimizing variation of feature means across operating environments. An additional environment-varying subspace, extracted from the same environment statistics, is used alongside the stable subspace for monitoring. We do not perform interventions or counterfactual reasoning, and marginal lagged covariance is weaker than Granger causality. The contributions of this work are as follows:

- We identify two complementary, causally motivated criteria for time-series features: temporal target relevance encoded in lagged covariances and cross-environment mean stability.
- We develop CFL, a generalized Rayleigh-quotient framework that couples a stable subspace with an environment-varying subspace and a slow-feature monitor, all used jointly for monitoring.
- We provide theoretical analysis establishing conditions under which CFL identifies a mean-invariant predictive subspace, together with a deterministic perturbation bound that states exactly what must be controlled in finite samples.

2. Related Work

2.1. Temporal Causality and Granger Causality

Granger causality (Granger, 1969) formalizes temporal predictive relationships: X Granger-causes Y if past values of X help predict Y beyond what past values of Y alone provide. This concept has been extended to multivariate settings (Lütkepohl, 2005) and nonlinear systems (Tank et al., 2022). Recent work on causal discovery from time series (Runge et al., 2019) has developed methods like PCMCI that combine constraint-based causal discovery with temporal structure.

In process monitoring, temporal effects are crucial because faults propagate through the process with characteristic delays determined by physical dynamics. Dynamic latent variable methods implicitly capture temporal predictability. Dynamic PCA (Ku et al., 1995) augments observations with lagged copies before applying PCA. DiPCA (Dong & Qin, 2018a) constructs time-lagged feature matrices and

maximizes covariance between current latent scores and predictions from past values:

$$\max_{\mathbf{w}} \text{Cov}(\mathbf{w}^\top \mathbf{x}_t, \hat{\mathbf{w}}^\top \mathbf{x}_{t-1:t-s}). \quad (1)$$

DiCCA (Dong & Qin, 2018b) similarly maximizes correlation:

$$\max_{\mathbf{w}} \text{Corr}(\mathbf{w}^\top \mathbf{x}_t, \hat{\mathbf{w}}^\top \mathbf{x}_{t-1:t-s}). \quad (2)$$

DiPLS (Dong & Qin, 2015) incorporates output variables, maximizing covariance between outputs and predicted latent scores. These methods capture how past states help predict current states but do not explicitly encourage invariance across operating conditions.

2.2. Causal Invariance and Independent Mechanisms

The principle of independent causal mechanisms (Peters et al., 2017; Schölkopf et al., 2012) motivates invariant causal prediction (ICP) (Peters et al., 2016) and invariant risk minimization (IRM) (Arjovsky et al., 2019), which identify causal features by finding representations whose predictive relationships are invariant across environments. Recent work has extended these ideas to heterogeneous and nonstationary data (Huang et al., 2020). In this paper, we focus on a tractable spectral formulation that enforces first-order (mean) invariance across environments.

Our construction has close relatives in classical multivariate analysis that we wish to make explicit. The environment matrix $\mathbf{A}_e$ coincides with the between-class scatter matrix of Fisher discriminant analysis (Fisher, 1936) when environments are treated as class labels, and our constrained trace formulation is the classical ratio-trace relaxation studied in the trace-ratio literature (Wang et al., 2007). Stationary Subspace Analysis (von Bünau et al., 2009) seeks projections whose first and second moments are stationary across time epochs; Domain-Invariant Component Analysis (Muandet et al., 2013) matches feature distributions across domains via kernel embeddings; and anchor regression (Rothenhäusler et al., 2021) interpolates between ordinary least squares and invariance using exogenous anchor variables. Relative to these, CFL couples an explicitly temporal target-relevance term (through lagged embeddings and one-step-ahead targets) with a first-order invariance penalty in a single generalized eigenproblem, and additionally extracts an environment-varying subspace that is used for monitoring.

Slow Feature Analysis (SFA) (Wiskott & Sejnowski, 2002) provides a complementary perspective. By extracting slowly varying features:

$$\min_{\mathbf{W}} \frac{\text{Tr}(\mathbf{W}^\top \boldsymbol{\Omega}_d \mathbf{W})}{\text{Tr}(\mathbf{W}^\top \boldsymbol{\Sigma}_d \mathbf{W})}, \quad (3)$$

where $\boldsymbol{\Omega}_d$ measures temporal variation rate and $\boldsymbol{\Sigma}_d$ measures variance, SFA assumes that meaningful mechanisms

vary slowly. While related in spirit, SFA enforces temporal smoothness rather than cross-environment invariance.

2.3. Process Monitoring and Fault Detection

Statistical process monitoring has a rich history (Chiang et al., 2000; Qin, 2012; Ge et al., 2013). PCA-based methods (Jackson, 1991; Nomikos & MacGregor, 1994) project data onto principal directions and monitor T^2 and SPE statistics. Extensions incorporate dynamics (Ku et al., 1995), quality variables (Dong & Qin, 2015), and nonlinear relationships (Kruger & Xie, 2012). However, these methods primarily capture statistical correlations without explicitly modeling causal structure, limiting robustness under distribution shift.

2.4. Bridging Temporal Relevance and Invariance

Our work bridges the temporal and invariance perspectives. We argue that robust causal features should satisfy both properties: (1) encode temporally predictive structure via time-lagged representations; and (2) be stable across environments, operationalized here as mean invariance. Existing methods optimize for one property but not both, potentially missing features that are temporally informative but environment-sensitive, or stable but not temporally informative.

3. Problem Formulation

3.1. Setting and Notation

Let $\mathbf{X} \in \mathbb{R}^{N \times m}$ denote time-series observations with N samples and m variables. We denote target variables (key process indicators) as $\mathbf{Y} \in \mathbb{R}^{N \times L}$ with components $y_{i,t}$. In our monitoring application the targets are L selected process measurements taken *one step ahead*: features computed from the past window at time t are trained to be predictive of $y_{i,t+1}$ (the concrete choice of targets is specified in Section 6).

To capture temporal structure, we construct time-lagged feature vectors:

$$\mathbf{z}_t = [\mathbf{x}_t^\top, \mathbf{x}_{t-1}^\top, \dots, \mathbf{x}_{t-s}^\top]^\top \in \mathbb{R}^{m(s+1)}, \quad (4)$$

where s is the lag order. Let $d = m(s+1)$ be the dimension of $\mathbf{z}_t$. We consider data from multiple operating environments $\mathcal{E} = \{e_1, \dots, e_K\}$, representing different operating modes, seasonal conditions, or process configurations.

3.2. Time-Lagged Covariance Structure

We define the (population) time-lagged covariance matrix

$$\Sigma_d \triangleq \text{Cov}(\mathbf{z}_t) = \mathbb{E}[(\mathbf{z}_t - \mathbb{E}[\mathbf{z}_t])(\mathbf{z}_t - \mathbb{E}[\mathbf{z}_t])^\top], \quad (5)$$

and its sample estimate

$$\hat{\Sigma}_d = \frac{1}{N-s-1} \sum_{t=s+1}^N (\mathbf{z}_t - \bar{\mathbf{z}})(\mathbf{z}_t - \bar{\mathbf{z}})^\top. \quad (6)$$

The matrix Σ_d is block-structured:

$$\Sigma_d = \begin{bmatrix} \text{Cov}(\mathbf{x}_t, \mathbf{x}_t) & \cdots & \text{Cov}(\mathbf{x}_t, \mathbf{x}_{t-s}) \\ \vdots & \ddots & \vdots \\ \text{Cov}(\mathbf{x}_{t-s}, \mathbf{x}_t) & \cdots & \text{Cov}(\mathbf{x}_{t-s}, \mathbf{x}_{t-s}) \end{bmatrix}. \quad (7)$$

The off-diagonal blocks $\text{Cov}(\mathbf{x}_t, \mathbf{x}_{t-k})$ encode lagged cross-dependencies. We also define the difference covariance matrix:

$$\begin{aligned} \Omega_d &\triangleq \text{Cov}(\mathbf{z}_t - \mathbf{z}_{t-1}) \\ &= \mathbb{E}[(\Delta \mathbf{z}_t - \mathbb{E}[\Delta \mathbf{z}_t])(\Delta \mathbf{z}_t - \mathbb{E}[\Delta \mathbf{z}_t])^\top], \end{aligned} \quad (8)$$

where $\Delta \mathbf{z}_t = \mathbf{z}_t - \mathbf{z}_{t-1}$, and its sample estimate

$$\hat{\Omega}_d = \frac{1}{N-s-2} \sum_{t=s+2}^N (\mathbf{z}_t - \mathbf{z}_{t-1})(\mathbf{z}_t - \mathbf{z}_{t-1})^\top. \quad (9)$$

3.3. Two Aspects of Causal Features

Definition 3.1 (Temporal Target Relevance). A feature direction $\mathbf{w}$ is temporally relevant for target component y_i if the time-lagged projection $\mathbf{w}^\top \mathbf{z}_t$ has non-zero covariance with the next-step target:

$$\text{Cov}(\mathbf{w}^\top \mathbf{z}_t, y_{i,t+1}) \neq 0. \quad (10)$$

This indicates that the combination of current and past values encoded in $\mathbf{w}^\top \mathbf{z}_t$ carries information about the future evolution of y_i .

Definition 3.2 (Environment Mean Invariance). A feature direction $\mathbf{w}$ is (first-order) environment-invariant if its mean is invariant across environments:

$$\mathbb{E}^e[\mathbf{w}^\top \mathbf{z}] = \mathbb{E}^{e'}[\mathbf{w}^\top \mathbf{z}] \quad \forall e, e' \in \mathcal{E}. \quad (11)$$

Equivalently, if $\mu_z^{(k)} \triangleq \mathbb{E}[\mathbf{z} \mid e = e_k]$, then $\mathbf{w}^\top \mu_z^{(k)}$ is constant over k .

CFL as defined in this paper enforces mean invariance via a between-environment mean-shift penalty. Stronger distributional invariance (e.g., $P^e(\mathbf{w}^\top \mathbf{z})$ identical across e) would require additional penalties such as higher-moment matching or kernel discrepancies. Our goal is to find a projection $\mathbf{W}_s$ that extracts features satisfying both properties: temporally relevant and environment mean-invariant.

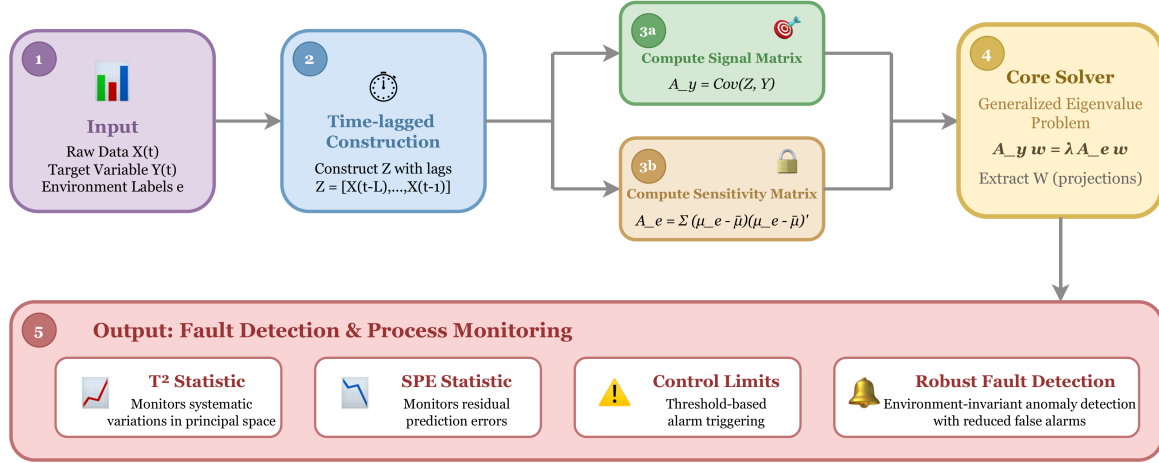


Figure 2. Framework of Causal Feature Learning via Generalized Rayleigh Quotients.

4. Methodology

4.1. Target-Relevance and Invariance Matrices

To capture temporal relevance to targets, define

$$\mathbf{A}_y = \sum_{i=1}^L \Sigma_{zy_i} \Sigma_{y_i y_i}^{-1} \Sigma_{y_i z}, \quad (12)$$

where $\Sigma_{zy_i} = \mathbb{E}[(z_t - \mathbb{E}[z_t])(y_{i,t+1} - \mathbb{E}[y_{i,t+1}])] \in \mathbb{R}^d$ and $\Sigma_{y_i y_i} = \text{Var}(y_i) = \sigma_{y_i}^2$. The matrix $\mathbf{A}_y$ aggregates directions that are linearly predictive of the next-step targets. Since each summand is rank one, $\text{rank}(\mathbf{A}_y) \leq L$: at most L directions carry target signal, and generalized eigenvectors beyond the rank of $\mathbf{A}_y$ have zero eigenvalue and are not identifiable. We therefore always take $d_s \leq L$. To penalize deviations from mean invariance, define

$$\mathbf{A}_e = \sum_{k=1}^K \pi_k (\mu_z^{(k)} - \bar{\mu}_z)(\mu_z^{(k)} - \bar{\mu}_z)^\top, \quad (13)$$

where $\mu_z^{(k)} = \mathbb{E}[z | e = e_k]$, $\bar{\mu}_z = \sum_k \pi_k \mu_z^{(k)}$, and π_k is the proportion of samples from environment e_k . Large projection onto $\mathbf{A}_e$ indicates environment-dependent mean shifts, suggesting spurious sensitivity.

4.2. Generalized Rayleigh-Quotient Formulation

Let $\mathbf{B} \triangleq \mathbf{A}_e + \epsilon \mathbf{I} \succ 0$ with $\epsilon > 0$. We extract d_s features by solving the constrained trace maximization

$$\max_{\mathbf{W}_s \in \mathbb{R}^{d \times d_s}} \text{Tr}(\mathbf{W}_s^\top \mathbf{A}_y \mathbf{W}_s) \quad \text{s.t.} \quad \mathbf{W}_s^\top \mathbf{B} \mathbf{W}_s = \mathbf{I}_{d_s}. \quad (14)$$

For $d_s = 1$, (14) reduces to the standard generalized Rayleigh quotient $\max_{\mathbf{w} \neq 0} \frac{\mathbf{w}^\top \mathbf{A}_y \mathbf{w}}{\mathbf{w}^\top \mathbf{B} \mathbf{w}}$.

The objective in (14) balances two complementary terms that reflect the dual nature of causal features. The signal term $\text{Tr}(\mathbf{W}_s^\top \mathbf{A}_y \mathbf{W}_s)$ measures how well the extracted features capture information predictive of the target variables, leveraging the time-lagged representation to encode temporal dependencies. Meanwhile, the invariance constraint $\mathbf{W}_s^\top \mathbf{B} \mathbf{W}_s = \mathbf{I}$ penalizes directions that exhibit large environment-dependent mean shifts through the matrix $\mathbf{A}_e$, with the regularization parameter ϵ ensuring that $\mathbf{B}$ remains positive definite.

Stationary points of (14) satisfy $\mathbf{A}_y \mathbf{W}_s = \mathbf{B} \mathbf{W}_s \boldsymbol{\Lambda}$ for a symmetric matrix of multipliers $\boldsymbol{\Lambda}$; because the objective and the constraint are invariant under $\mathbf{B}$ -orthogonal rotations of $\mathbf{W}_s$, we may choose the basis in which $\boldsymbol{\Lambda}$ is diagonal, reducing stationarity to the generalized eigenvalue problem

$$\mathbf{A}_y \mathbf{w} = \lambda \mathbf{B} \mathbf{w}. \quad (15)$$

Let $\{\mathbf{w}_i\}_{i=1}^{d_s}$ be the generalized eigenvectors associated with the largest eigenvalues $\lambda_1 \geq \dots \geq \lambda_{d_s}$, normalized such that $\mathbf{W}_s^\top \mathbf{B} \mathbf{W}_s = \mathbf{I}_{d_s}$. Then $\mathbf{W}_s = [\mathbf{w}_1, \dots, \mathbf{w}_{d_s}]$ attains the global maximum of (14). We note that (14) is the *ratio-trace* (constrained-trace) formulation, whose global optimum is exactly the generalized eigenvector solution; the alternative *trace-ratio* objective $\text{Tr}(\mathbf{W}^\top \mathbf{A}_y \mathbf{W}) / \text{Tr}(\mathbf{W}^\top \mathbf{B} \mathbf{W})$ generally requires iterative algorithms (Wang et al., 2007) and is not used here.

4.3. Connection to Existing Methods

Slow Feature Analysis. SFA can be written in the standard constrained form

$$\min_{\mathbf{W}} \text{Tr}(\mathbf{W}^\top \boldsymbol{\Omega}_d \mathbf{W}) \quad \text{s.t.} \quad \mathbf{W}^\top \boldsymbol{\Sigma}_d \mathbf{W} = \mathbf{I}. \quad (16)$$

Here, Ω_d captures temporal variation via differences Δz_t , and minimizing it extracts slowly varying features. The constraint Σ_d provides variance normalization that prevents degenerate solutions. While SFA uses temporal smoothness as a proxy for invariance, CFL explicitly penalizes cross-environment mean shifts via A_e . This distinction matters when informative causal signals exhibit transients (e.g., during fault propagation) that SFA would suppress.

Dynamic Latent Variable Methods. DiPCA and DiCCA emphasize temporal predictability but do not explicitly control environment sensitivity; a direction may be strongly predictable yet driven by environment-dependent mean shifts. DiPLS incorporates target variables but maximizes predictive covariance without an explicit invariance penalty. In contrast, CFL balances target relevance (via A_y) and mean invariance (via A_e), yielding features that are both temporally informative and robust to environment-dependent mean shifts.

4.4. Environment-Varying Subspace

The stable subspace deliberately suppresses directions whose means shift across environments. For *monitoring*, however, precisely those directions are informative: faults and operating-condition changes manifest as shifts along them. We therefore also extract an additional *varying* subspace by maximizing environment sensitivity relative to total variance,

$$\begin{aligned} \max_{\mathbf{W}_v \in \mathbb{R}^{d \times d_v}} \quad & \text{Tr}(\mathbf{W}_v^\top \mathbf{A}_e \mathbf{W}_v) \\ \text{s.t.} \quad & \mathbf{W}_v^\top (\Sigma_d + \epsilon \mathbf{I}) \mathbf{W}_v = \mathbf{I}_{d_v}, \end{aligned} \quad (17)$$

solved by the top d_v generalized eigenvectors of $(\mathbf{A}_e, \Sigma_d + \epsilon \mathbf{I})$. Since $\text{rank}(\mathbf{A}_e) \leq K - 1$ (Lemma 5.3), we take $d_v \leq K - 1$. As the ablations in Section 6 show, monitoring this subspace is essential for detection performance.

4.5. Fault Detection Based on CFL Features

Given the learned subspaces, define scores $\mathbf{s}_t = \mathbf{W}_s^\top \mathbf{z}_t$ and $\mathbf{u}_t = \mathbf{W}_v^\top \mathbf{z}_t$.

T^2 Statistics. Let $\hat{\Sigma}_s$ and $\hat{\Sigma}_u$ be the sample covariances of $\{\mathbf{s}_t\}$ and $\{\mathbf{u}_t\}$ on normal training data. We monitor

$$T_{s,t}^2 = \mathbf{s}_t^\top \hat{\Sigma}_s^{-1} \mathbf{s}_t, \quad T_{v,t}^2 = \mathbf{u}_t^\top \hat{\Sigma}_u^{-1} \mathbf{u}_t. \quad (18)$$

SPE Statistic. Let $\mathbf{W} = [\mathbf{W}_s, \mathbf{W}_v]$ and let $\mathbf{P}_\mathbf{W} = \mathbf{W} \mathbf{W}^+$ denote the orthogonal projector onto $\text{span}(\mathbf{W})$, where $\mathbf{W}^+$ is the Moore–Penrose pseudoinverse. We monitor the residual energy

$$\text{SPE}_t = \|(\mathbf{I} - \mathbf{P}_\mathbf{W}) \mathbf{z}_t\|^2. \quad (19)$$

Slowness Monitor. Some faults perturb primarily the *dynamics* of the process—how fast trajectories move—while

Algorithm 1 Causal Feature Learning

Require: Time-series $\mathbf{X} \in \mathbb{R}^{N \times m}$, target indices defining $\mathbf{Y}$ (L columns, one step ahead), environment segmentation $e \in \{e_1, \dots, e_K\}$, lag order s , dimensions $d_s \leq L$ and $d_v \leq K - 1$, regularization ϵ , false-alarm budget α , normal validation sequence

- 1: Construct time-lagged matrix $\mathbf{Z}$ (rows are $\mathbf{z}_t$) and aligned targets $y_{i,t+1}$
 - 2: Compute sample estimates $\hat{\mathbf{A}}_y$ (Eq. 12), $\hat{\mathbf{A}}_e$ (Eq. 13), and $\hat{\Sigma}_d$
 - 3: Solve $\hat{\mathbf{A}}_y \mathbf{w} = \lambda(\hat{\mathbf{A}}_e + \epsilon \mathbf{I}) \mathbf{w}$; collect top d_s eigenvectors as $\mathbf{W}_s$
 - 4: Solve $\hat{\mathbf{A}}_e \mathbf{w} = \lambda(\hat{\Sigma}_d + \epsilon \mathbf{I}) \mathbf{w}$; collect top d_v eigenvectors as $\mathbf{W}_v$
 - 5: Solve the slow-feature eigenproblem (Eq. 16) on standardized $\mathbf{x}_t$; collect the M slowest directions as $\mathbf{W}_{\text{sf}}$
 - 6: Compute training score covariances $\hat{\Sigma}_s$, $\hat{\Sigma}_u$ and the projector $\mathbf{P}_\mathbf{W}$
 - 7: Select a shared marginal rank q for T_s^2 , T_v^2 , T_{sf}^2 , and SPE so their validation OR-alarm rate is at most α
 - 8: **return** $\mathbf{W}_s$, $\mathbf{W}_v$, $\mathbf{W}_{\text{sf}}$, score covariances, thresholds
-

leaving the mean structure captured by the subspaces above nearly unchanged (Fault 19 of the TEP benchmark is the canonical example). We therefore add a fourth statistic built from a third generalized eigenproblem, the slow-feature problem of Eq. (16) applied to the standardized instantaneous measurements $\mathbf{x}_t$: after whitening, let $\mathbf{W}_{\text{sf}}$ collect the M slowest directions of the whitened difference covariance, with M set by the standard SFA rule that retains directions slower than the fastest raw variable (Wiskott & Sejnowski, 2002). We monitor the Hotelling statistic of the slow features,

$$T_{\text{sf},t}^2 = \|\mathbf{W}_{\text{sf}}^\top \mathbf{x}_t\|^2, \quad (20)$$

which coincides with SFA’s slow-feature T^2 statistic. Including it makes the complementarity of target relevance, environment sensitivity, and slowness explicit within a single jointly calibrated monitor set; the “no slowness” ablation in Section 6.3 isolates its contribution.

The complete CFL framework is illustrated in Figure 2, and as described in Algorithm 1, CFL extracts features by solving three generalized eigenvalue problems: two on the lagged embedding (stable and varying subspaces) and the slow-feature problem on the instantaneous measurements.

5. Theoretical Analysis

5.1. Causal/Spurious Feature Decomposition

Assumption 5.1 (Predictive, Spurious, and Residual Subspaces). There exists a direct-sum decomposition $\mathbb{R}^d = \mathcal{C} \oplus \mathcal{S} \oplus \mathcal{R}$ such that $\mathbf{z} = \mathbf{z}_c + \mathbf{z}_s + \mathbf{z}_r$ with $\mathbf{z}_c \in \mathcal{C}$,

$z_s \in \mathcal{S}$, $z_r \in \mathcal{R}$, where: (i) the *predictive invariant* subspace $\mathcal{C}$ is mean-invariant, $\mathcal{C} \subseteq \ker(\mathbf{A}_e)$, and predictively rich: there exists $\gamma_c > 0$ with $\mathbf{w}^\top \mathbf{A}_y \mathbf{w} \geq \gamma_c \|\mathbf{w}\|^2$ for all $\mathbf{w} \in \mathcal{C}$; (ii) the *spurious mean-shift* subspace $\mathcal{S}$ is environment-diverse: there exists $\sigma_s > 0$ with $\mathbf{w}^\top \mathbf{A}_e \mathbf{w} \geq \sigma_s \|\mathbf{w}\|^2$ for all $\mathbf{w} \in \mathcal{S}$; (iii) the *residual* subspace $\mathcal{R} \subseteq \ker(\mathbf{A}_e) \cap \ker(\mathbf{A}_y)$ is neither target-predictive nor mean-shifted. By Lemma 5.3, these conditions entail $\dim(\mathcal{C}) \leq L$ and $\dim(\mathcal{S}) \leq K - 1$; the residual subspace absorbs the remaining $d - \dim(\mathcal{C}) - \dim(\mathcal{S})$ dimensions.

Remark 5.2 (Scope of the decomposition). The three-way decomposition is essential in the regime relevant to our experiments: with lagged dimension $d = m(s + 1)$ in the hundreds, but L targets and K environments both on the order of ten, the residual subspace is by far the largest ($\dim(\mathcal{R}) \geq d - L - (K - 1)$). A two-way decomposition $\mathbb{R}^d = \mathcal{C} \oplus \mathcal{S}$ with the coercivity conditions above would force $d \leq L + K - 1$ and thus could not hold in this regime. In our TEP configuration ($m = 52$, $s = 5$, $d = 312$, $L = 12$, $K = 10$), $\dim(\mathcal{C}) \leq 12$, $\dim(\mathcal{S}) \leq 9$, and $\dim(\mathcal{R}) \geq 291$.

5.2. Properties of Key Matrices

Lemma 5.3 (Properties of $\mathbf{A}_y$ and $\mathbf{A}_e$). *The target-relevance matrix $\mathbf{A}_y$ defined in (12) is positive semidefinite with $\text{rank}(\mathbf{A}_y) \leq L$, and for any $\mathbf{w}$ we have $\mathbf{w}^\top \mathbf{A}_y \mathbf{w} = \sum_{i=1}^L (\mathbf{w}^\top \Sigma_{zy_i})^2 / \sigma_{y_i}^2$. The environment sensitivity matrix $\mathbf{A}_e$ defined in (13) is positive semidefinite with $\text{rank}(\mathbf{A}_e) \leq K - 1$, and satisfies the law of total variance $\text{Var}(\mathbf{z}) = \mathbb{E}[\text{Var}(\mathbf{z} | e)] + \mathbf{A}_e$. Moreover, for any mean-invariant direction $\mathbf{w}$ satisfying $\mathbf{w}^\top \mu_z^{(k)}$ constant over k , we have $\mathbf{w}^\top \mathbf{A}_e \mathbf{w} = 0$.*

Proof Sketch. For $\mathbf{A}_y$, expanding $\mathbf{w}^\top \mathbf{A}_y \mathbf{w}$ yields a sum of squared terms, establishing positive semidefiniteness; each summand is rank-1, so $\text{rank}(\mathbf{A}_y) \leq L$. For $\mathbf{A}_e$, positive semidefiniteness follows from its form as a weighted sum of outer products; the rank bound follows from the constraint $\sum_k \pi_k (\mu_z^{(k)} - \bar{\mu}_z) = \mathbf{0}$. See Appendix A.1 for details. $\square$

5.3. Identification of Mean-Invariant Predictive Subspace

Theorem 5.4 (Mean-Invariant Predictive Subspace Identification). *Let $\mathbf{B} = \mathbf{A}_e + \epsilon \mathbf{I} \succ 0$ and suppose Assumption 5.1 holds with $r_c = \dim(\mathcal{C})$. Then, as $\epsilon \rightarrow 0$, exactly the top r_c generalized eigenvalues of $\mathbf{A}_y \mathbf{w} = \lambda(\mathbf{A}_e + \epsilon \mathbf{I}) \mathbf{w}$ diverge on the order of $1/\epsilon$, while the remaining eigenvalues are bounded by $\|\mathbf{A}_y\|_2 / \sigma_s$ uniformly in ϵ ; every direction in $\mathcal{R}$ is a generalized eigenvector with eigenvalue zero. If, in addition, $\mathcal{C} \perp \mathcal{R}$, the top- r_c generalized eigenspace converges to $\mathcal{C}$.*

Proof Sketch. Define $R(\mathbf{w}) = \mathbf{w}^\top \mathbf{A}_y \mathbf{w} / \mathbf{w}^\top \mathbf{B} \mathbf{w}$. For $\mathbf{w} \in \mathcal{C} \setminus \{0\}$, $\mathcal{C} \subseteq \ker(\mathbf{A}_e)$ implies $\mathbf{w}^\top \mathbf{A}_e \mathbf{w} = 0$, so $R(\mathbf{w}) \geq \gamma_c / \epsilon \rightarrow \infty$; for $\mathbf{w} \in \mathcal{S} \oplus \mathcal{R}$, $R(\mathbf{w}) \leq \|\mathbf{A}_y\|_2 / \sigma_s$; and $R \equiv 0$ on $\mathcal{R}$. The Courant–Fischer min–max theorem then yields the spectral separation. The convergence claim follows from the eigenvalue equation itself, not from Rayleigh quotients: any eigenvector with $\lambda \neq 0$ satisfies $\mathbf{w} = (\mathbf{A}_y \mathbf{w} - \lambda \mathbf{A}_e \mathbf{w}) / (\lambda \epsilon) \in \text{range}(\mathbf{A}_y) + \text{range}(\mathbf{A}_e) \subseteq \mathcal{R}^\perp$, and coercivity on $\mathcal{S}$ forces the $\mathcal{S}$ -component of a top eigenvector to vanish at rate $O(\sqrt{\epsilon})$; under $\mathcal{C} \perp \mathcal{R}$ this yields convergence to $\mathcal{C}$. See Appendix A.2 for the complete argument; without $\mathcal{C} \perp \mathcal{R}$ the limit subspace is $(\mathcal{C} \oplus \mathcal{R}) \cap \mathcal{R}^\perp$, whose deviation from $\mathcal{C}$ is governed by the principal angles between $\mathcal{C}$ and $\mathcal{R}$. $\square$

Corollary 5.5 (Spectral Gap). *Under the conditions of Theorem 5.4, if additionally $\lambda_{r_c+1} > 0$, then for ϵ sufficiently small,*

$$\frac{\lambda_{r_c}}{\lambda_{r_c+1}} \geq \frac{\gamma_c \sigma_s}{\epsilon \|\mathbf{A}_y\|_2}; \quad (21)$$

when $\lambda_{r_c+1} = 0$ (e.g., all non-divergent directions lie in $\mathcal{R}$), the separation between divergent and non-divergent eigenvalues is unbounded and the statement holds trivially with the gap measured by the difference $\lambda_{r_c} - \lambda_{r_c+1} \geq \gamma_c / \epsilon$.

Proof Sketch. From Theorem 5.4, $\lambda_{r_c} \geq \gamma_c / \epsilon$, and the min–max argument in Appendix A.2 gives $\lambda_{r_c+1} \leq \|\mathbf{A}_y\|_2 / \sigma_s$ unconditionally. The ratio follows immediately. See Appendix A.3 for details. $\square$

5.4. Finite-Sample Perturbation

Let $\mathbf{B} = \mathbf{A}_e + \epsilon \mathbf{I}$ and $\mathbf{M} = \mathbf{B}^{-1/2} \mathbf{A}_y \mathbf{B}^{-1/2}$, and define their sample analogues $\hat{\mathbf{B}}$ and $\hat{\mathbf{M}}$.

Proposition 5.6 (Deterministic whitened-subspace stability). *Let $\mathbf{V}_{d_s}$ and $\hat{\mathbf{V}}_{d_s}$ contain the top d_s eigenvectors of $\mathbf{M}$ and $\hat{\mathbf{M}}$, respectively. If $\lambda_{d_s}(\mathbf{M}) - \lambda_{d_s+1}(\mathbf{M}) \geq \Delta > 0$ and $\eta = \|\hat{\mathbf{M}} - \mathbf{M}\|_2 < \Delta/2$, then*

$$\|\sin \Theta(\hat{\mathbf{V}}_{d_s}, \mathbf{V}_{d_s})\|_2 \leq \frac{2\eta}{\Delta}. \quad (22)$$

Proof. This is the Davis–Kahan $\sin \Theta$ bound applied to the two symmetric whitened matrices; the condition $\eta < \Delta/2$ preserves separation of the target eigenvalue cluster. $\square$

Remark 5.7. Proposition 5.6 is deterministic: it separates the spectral sensitivity of CFL from the statistical problem of bounding η . A sample-size rate for overlapping lagged vectors requires a dependence inequality whose tail, boundedness, and mixing assumptions match the data. In particular, the matrix Bernstein result of Banna et al. (2016) assumes uniformly bounded, geometrically absolutely regular matrices and does not by itself justify a sub-Gaussian rate under merely summable β -mixing coefficients. We therefore make

no unsupported N -dependent rate claim. The factor $B^{-1/2}$ also explains why very small ϵ can amplify estimation error; the ϵ ablation in Section 6.3 evaluates this sensitivity empirically. Mapping the whitened eigenspace back to generalized directions additionally requires controlling the perturbation of $B^{-1/2}$.

5.5. Relation to Causal Principles, and Limitations

Proposition 5.8 (Temporal Relevance via Lagged Cross-Covariance). *Let $e_{j,k} \in \mathbb{R}^d$ be the canonical basis vector corresponding to $X_{j,t-k}$ in $\mathbf{z}_t$. If $\text{Cov}(X_{j,t-k}, y_{i,t+1}) \neq 0$ for some target y_i , then $e_{j,k}^\top \mathbf{A}_y e_{j,k} > 0$.*

Proof Sketch. A nonzero covariance implies $[\Sigma_{zy_i}]_{j+km} \neq 0$. Since $e_{j,k}^\top \mathbf{A}_y e_{j,k} = \sum_{\ell=1}^L [\Sigma_{zy_\ell}]_{j+km}^2 / \sigma_{y_\ell}^2$, the term for $\ell = i$ is strictly positive. See Appendix A.4 for details. $\square$

We emphasize that Proposition 5.8 concerns *marginal* lagged association, which is strictly weaker than Granger causality: Granger causality requires predictive content conditional on the target’s own past (Granger, 1969), and a nonzero marginal lagged covariance can arise from confounding or from the target’s autocorrelation alone. $\mathbf{A}_y$ should therefore be read as a target-relevance matrix, not as evidence of a causal link.

Proposition 5.9 (Invariance of the Conditional Mechanism). *Assume a structural equation model $\mathbf{Y} = f(\mathbf{z}_c) + \boldsymbol{\eta}$ holds across environments, where f and the distribution of $\boldsymbol{\eta}$ do not depend on e , and $(\mathbf{z}_c, \boldsymbol{\eta}) \perp\!\!\!\perp e$. Assume further that either the decomposition of Assumption 5.1 is orthogonal ($\mathcal{S} \oplus \mathcal{R} = \mathcal{C}^\perp$) and $\mathbf{W}_s$ spans $\mathcal{C}$ with $\mathbf{W}_s^\top \mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in \mathcal{S} \oplus \mathcal{R}$, or that $\mathbf{W}_s$ is constructed to annihilate $\mathcal{S} \oplus \mathcal{R}$. Then $\mathbf{s} = \mathbf{W}_s^\top \mathbf{z}$ is a function of $\mathbf{z}_c$ alone, and $P^e(\mathbf{Y} | \mathbf{s}) = P^{e'}(\mathbf{Y} | \mathbf{s})$ for all $e, e' \in \mathcal{E}$.*

Proof Sketch. The annihilation condition is what makes $\mathbf{s} = \mathbf{W}_s^\top (\mathbf{z}_c + \mathbf{z}_s + \mathbf{z}_r) = \mathbf{W}_s^\top \mathbf{z}_c$ a function of $\mathbf{z}_c$ alone; without it, $\mathbf{s}$ mixes in components of $\mathbf{z}_s$ and the conclusion fails in general. The pair $(\mathbf{Y}, \mathbf{s}) = (f(\mathbf{z}_c) + \boldsymbol{\eta}, \mathbf{W}_s^\top \mathbf{z}_c)$ is then a measurable function of $(\mathbf{z}_c, \boldsymbol{\eta})$; since $(\mathbf{z}_c, \boldsymbol{\eta}) \perp\!\!\!\perp e$ and neither f nor $P(\boldsymbol{\eta})$ depends on e , the joint law of $(\mathbf{Y}, \mathbf{s})$ is identical across environments, and hence so is the conditional law $P^e(\mathbf{Y} | \mathbf{s})$. No injectivity of $\mathbf{W}_s$ on $\mathcal{C}$ is required. See Appendix A.5 for details. $\square$

Limitations of the causal interpretation. CFL optimizes two proxy criteria—lagged target relevance and first-order (mean) invariance—that are motivated by, but not equivalent to, causal structure. Mean invariance is neither necessary nor sufficient for causal validity: a causal mechanism may shift in higher moments while keeping means constant, and a spurious association can happen to have environment-invariant means. Proposition 5.9 makes the gap explicit: the

causal-style conclusion holds only *given* a structural model in which the invariant subspace coincides with the causal input. Moreover, when environments are constructed as temporal segments of data from a single operating mode (as in our TEP experiments), $\mathbf{A}_e$ captures slow drift rather than genuine interventional heterogeneity. We therefore regard CFL as an invariance-regularized predictive feature extractor whose causal interpretation strengthens with the quality of the available environment structure.

Proposition 5.10 (Complexity Analysis). *The computational complexity of CFL is $O(Nd^2 + d^3)$, where $d = m(s+1)$. For $N \gg d$, the cost is dominated by covariance estimation.*

Proof Sketch. Time-lagged matrix construction requires $O(Nms)$, covariance estimation requires $O(Nd^2)$, and the generalized eigenvalue problem requires $O(d^3)$. Summing these yields the total complexity. See Appendix A.6 for the breakdown. $\square$

6. Experiments

6.1. Experimental Setup

Data. We evaluate CFL on the Tennessee Eastman Process (TEP) (Downs & Vogel, 1993), using the widely adopted simulation data of Chiang et al. (2000). The process comprises $m = 52$ variables: 41 process measurements (XMEAS 1–41) and 11 manipulated variables (XMV 1–11), generated from interconnected reactor, separator, and stripper units. Training uses the normal-operation run (d00, 500 samples). A second, independent normal-operation run (d00_te, 960 samples) is used exclusively to calibrate alarm thresholds. Testing uses the 21 fault runs (d01_te–d21_te), each containing 960 samples with the fault introduced at sample 160 (samples are 3 minutes apart). All data are standardized using training-set statistics.

Targets and environments. The $L = 12$ target variables defining $\mathbf{A}_y$ are XMEAS 7–10 (reactor pressure, level, temperature, purge rate), XMEAS 18–22 (stripper temperature and flows, compressor work, cooling-water outlet temperatures), and XMEAS 37–39 (product analyzer compositions), taken one step ahead of the lagged embedding. These are key downstream indicators of process state. Because the training data come from a single operating mode, no genuine environment labels exist; we construct $K = 10$ proxy environments by segmenting the training embedding into contiguous windows of near-equal length, so that $\mathbf{A}_e$ captures slow within-mode drift. We point this explicitly as a limitation: window-based environments provide much weaker heterogeneity than true operating-mode changes.

CFL configuration. We use lag order $s = 5$ (so $d = m(s+1) = 312$), stable dimension $d_s = L = 12$ and vary-

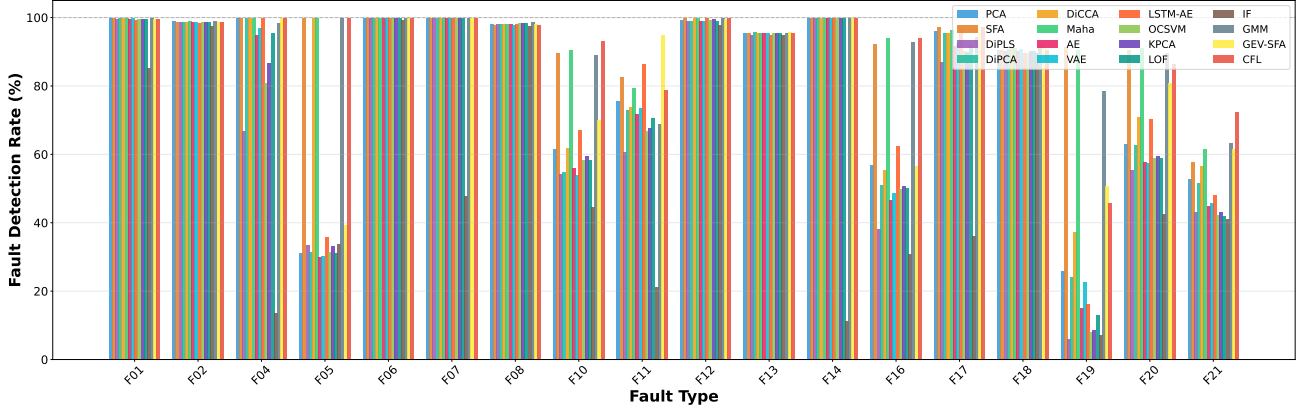


Figure 3. Per-fault FDR for all 16 methods on the 18 detectable TEP faults. Faults 03, 09, and 15 are omitted from this figure because they are essentially undetectable for every method at this false-alarm level; Table 1 reports the same 18 faults together with realized FAR and detection delay.

ing dimension $d_v = K - 1 = 9$ (both at their identifiable maxima; cf. Lemma 5.3), and $\epsilon = 10^{-4}$. No hyperparameter was tuned on the fault runs.

Evaluation protocol. Every method is trained on the same training run and calibrated on the same validation run using the joint empirical OR procedure described above. We report per-fault fault detection rate (FDR, alarm rate on raw samples ≥ 160), false-alarm rate (FAR, alarm rate on the common raw-sample window 10–149), and detection delay (samples from fault onset to the first run of three consecutive alarms). Raw samples 0–9 are excluded uniformly as startup, and 150–159 form a buffer before onset. Score offsets introduced by lagged or windowed models are explicitly mapped back to raw sample indices. Reporting FDR together with FAR is essential: FDR alone can be inflated arbitrarily by an ill-calibrated threshold. All reported averages are computed over the 18 detectable faults; Faults 03, 09, and 15 are excluded throughout, as they are near-undetectable for every method at this false-alarm level. All experiments use fixed random seeds; the complete pipeline and machine-readable outputs are retained in the reproducibility archive.

Baselines. We compare against 15 unsupervised baselines in three groups. (i) Dynamic latent-variable methods: PCA (Jackson, 1991), SFA (Wiskott & Sejnowski, 2002), DiPLS (Dong & Qin, 2015), DiPCA (Dong & Qin, 2018a), and DiCCA (Dong & Qin, 2018b). (ii) Distance, density, and reconstruction methods: Mahalanobis distance (Mahalanobis, 1936), autoencoder (Hinton & Salakhutdinov, 2006), variational autoencoder (Kingma & Welling, 2014), LSTM autoencoder, one-class SVM (Schölkopf et al., 2001), kernel PCA (Schölkopf et al., 1998), local outlier factor (Breunig et al., 2000), isolation forest (Liu et al., 2008), and a Gaussian mixture model. (iii) GEV-SFA: a slowness-based generalized-eigenvalue method that solves $\text{eig}(\Sigma_d, \Omega_d + \epsilon \Sigma_d^+)$ on the same lagged embedding and

monitors T^2/SPE ; this was the detection engine used in an earlier version of this work and is included for transparency. We do not include supervised classifiers: fitting them requires labeled fault data, which a fair comparison against purely normal-data-trained monitors cannot provide.

6.2. Main Results

Table 1 and Figure 3 summarize the comparison. CFL attains the highest average FDR of the 16 methods (94.6% at 1.3% realized FAR; Mahalanobis 93.7%, SFA 93.5%, GMM 92.2%) together with the lowest mean detection delay (24 samples; next best: GEV-SFA and Mahalanobis at 26). The top margin is small, however: the paired CFL-minus-Mahalanobis FDR difference is +0.90 points with a 95% bootstrap interval of $[-0.13, 2.47]$, so the ordering at the top should be treated as descriptive rather than statistically resolved. CFL is best on 7 faults, in a three-way tie with SFA and Mahalanobis. A single design element accounts for most of the improvement over the three-statistic variant: the slowness monitor raises Fault 19—a dynamics-dominated fault to which the lagged-subspace statistics are nearly blind—from 45.8% to 90.0%, on par with SFA’s 91.4%.

Two observations deserve emphasis. First, Faults 03, 09, and 15 are essentially undetectable for every method at the 5% false-alarm budget, consistent with the long-standing consensus in the TEP literature. Apparent high detection rates on these faults reported under FDR-only protocols typically reflect uncontrolled false-alarm rates rather than genuine detectability. Second, the calibrated protocol reorders methods substantially relative to an FDR-only comparison: methods whose nominal parametric control limits are miscalibrated on this data benefit heavily from FDR-only evaluation, which is precisely why we calibrate all thresholds on held-out normal data.

6.3. Ablation Studies

Table 2 ablates each component and hyper-parameter of CFL under the identical calibrated protocol, reporting average FDR and realized FAR over the 18 detectable faults.

Temporal depth helps: the FDR rises from 92.5% without the lagged embedding ($s=0$) to 94.6% at the default $s=5$ and 95.0% at $s=8$, while the realized FAR falls from 2.1% to 1.2%. The stable dimension has little effect, as $d_s \in \{4, 8, 12\}$ varies the FDR only between 94.5% and 94.6%. The varying monitor matters more: removing it ($d_v=0$) lowers the FDR to 92.1%, and the intermediate $d_v=5$ recovers only part of the gap (92.9%).

The number of environments has a mild effect: $K=3$ attains a higher FDR (94.8%) but at a 3.2% realized FAR, and $K=20$ gives 94.0% at 2.1%, so the default $K=10$ offers the best balance (94.6% at 1.3%). Regularization is uncritical, with ϵ from 10^{-5} to 10^{-2} varying the FDR only between 93.8% and 94.6%.

Among the component variants, removing A_e costs one FDR point (94.6% $\rightarrow$ 93.6%) and doubles the realized FAR (1.3% $\rightarrow$ 2.6%). Removing A_y leaves the FDR unchanged (94.6%) but also doubles the realized FAR, so the same detection level is reached only at a worse false-alarm level. Removing the slowness monitor T_{sf}^2 causes the largest loss, lowering the FDR to 91.7%. Among these component variants, only the full model combines the highest FDR with the lowest realized FAR (94.6% at 1.3%).

7. Conclusion

This work addressed the challenge of extracting robust causal features from time-series data under distribution shifts. We found that jointly optimizing temporal target relevance and environment mean invariance yields monitoring features that outperform methods focusing on only one aspect. CFL provides a principled spectral framework based on generalized Rayleigh quotients, with theoretical guarantees for identifying mean-invariant predictive subspaces. The results demonstrate that incorporating causal invariance principles into feature learning can yield the most favorable detection–false-alarm operating point among all compared methods. Future work includes automatic target discovery, stronger distributional invariance via kernel-based penalties, and nonlinear extensions using neural networks.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, and more specifically, the theoretical foundations of combining temporal causality and invariance principles in feature learning. There are many potential societal consequences of our work, none which we feel

must be specifically highlighted here.

References

- Arjovsky, M., Bottou, L., Gulrajani, I., and Lopez-Paz, D. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019.
- Banna, M., Merlevède, F., and Youssef, P. Bernstein-type inequality for a class of dependent random matrices. *Random Matrices: Theory and Applications*, 5(2):1650006, 2016.
- Breunig, M. M., Kriegel, H.-P., Ng, R. T., and Sander, J. LOF: Identifying density-based local outliers. In *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, pp. 93–104, 2000.
- Chiang, L. H., Russell, E. L., and Braatz, R. D. *Fault Detection and Diagnosis in Industrial Systems*. Springer, 2000.
- Dong, Y. and Qin, S. J. Dynamic-inner partial least squares for dynamic data modeling. *IFAC-PapersOnLine*, 48: 117–122, 2015.
- Dong, Y. and Qin, S. J. A novel dynamic PCA algorithm for dynamic data modeling and process monitoring. *Journal of Process Control*, 67:1–11, 2018a.
- Dong, Y. and Qin, S. J. Dynamic latent variable analytics for process operations and control. *Computers & Chemical Engineering*, 114:69–80, 2018b.
- Downs, J. J. and Vogel, E. F. A plant-wide industrial process control problem. *Computers & Chemical Engineering*, 17(3):245–255, 1993.
- Fisher, R. A. The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2):179–188, 1936.
- Ge, Z., Song, Z., and Gao, F. Review of recent research on data-based process monitoring. *Industrial & Engineering Chemistry Research*, 52(10):3543–3562, 2013.
- Granger, C. W. J. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438, 1969.
- Hinton, G. E. and Salakhutdinov, R. R. Reducing the dimensionality of data with neural networks. *Science*, 313(5786):504–507, 2006.
- Huang, B., Zhang, K., Zhang, J., Ramsey, J., Sanchez-Romero, R., Glymour, C., and Schölkopf, B. Causal discovery from heterogeneous/nonstationary data. *Journal of Machine Learning Research*, 21(89):1–53, 2020.

- Jackson, J. E. *A User's Guide to Principal Components*. John Wiley & Sons, 1991.
- Kingma, D. P. and Welling, M. Auto-encoding variational Bayes. In *International Conference on Learning Representations*, 2014.
- Kruger, U. and Xie, L. *Statistical Monitoring of Complex Multivariate Processes: With Applications in Industrial Process Control*. John Wiley & Sons, 2012.
- Ku, W., Storer, R. H., and Georgakis, C. Disturbance detection and isolation by dynamic principal component analysis. *Chemometrics and Intelligent Laboratory Systems*, 30(1):179–196, 1995.
- Liu, F. T., Ting, K. M., and Zhou, Z.-H. Isolation forest. In *Proceedings of the 8th IEEE International Conference on Data Mining*, pp. 413–422, 2008.
- Lütkepohl, H. *New Introduction to Multiple Time Series Analysis*. Springer, 2005.
- Mahalanobis, P. C. On the generalized distance in statistics. *Proceedings of the National Institute of Sciences of India*, 2(1):49–55, 1936.
- Muandet, K., Balduzzi, D., and Schölkopf, B. Domain generalization via invariant feature representation. In *Proceedings of the 30th International Conference on Machine Learning*, pp. 10–18, 2013.
- Nomikos, P. and MacGregor, J. F. Monitoring batch processes using multiway principal component analysis. *AIChE Journal*, 40(8):1361–1375, 1994.
- Pearl, J. *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2nd edition, 2009.
- Peters, J., Bühlmann, P., and Meinshausen, N. Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society: Series B*, 78(5):947–1012, 2016.
- Peters, J., Janzing, D., and Schölkopf, B. *Elements of Causal Inference: Foundations and Learning Algorithms*. MIT Press, 2017.
- Qin, S. J. Survey on data-driven industrial process monitoring and diagnosis. *Annual Reviews in Control*, 36(2): 220–234, 2012.
- Rothenhäusler, D., Meinshausen, N., Bühlmann, P., and Peters, J. Anchor regression: Heterogeneous data meet causality. *Journal of the Royal Statistical Society: Series B*, 83(2):215–246, 2021.
- Runge, J., Bathiany, S., Bollt, E., Camps-Valls, G., Coumou, D., Deyle, E., Glymour, C., Kretschmer, M., Mahecha, M. D., Muñoz-Marí, J., van Nes, E. H., Peters, J., Quax, R., Reichstein, M., Scheffer, M., Schölkopf, B., Spirtes, P., Sugihara, G., Sun, J., Zhang, K., and Zscheischler, J. Inferring causation from time series in Earth system sciences. *Nature Communications*, 10(1):2553, 2019.
- Schölkopf, B., Smola, A., and Müller, K.-R. Nonlinear component analysis as a kernel eigenvalue problem. *Neural Computation*, 10(5):1299–1319, 1998.
- Schölkopf, B., Platt, J. C., Shawe-Taylor, J., Smola, A. J., and Williamson, R. C. Estimating the support of a high-dimensional distribution. *Neural Computation*, 13(7): 1443–1471, 2001.
- Schölkopf, B., Janzing, D., Peters, J., Sgouritsa, E., Zhang, K., and Mooij, J. M. On causal and anticausal learning. In *International Conference on Machine Learning*, pp. 1255–1262, 2012.
- Spirtes, P., Glymour, C., and Scheines, R. *Causation, Prediction, and Search*. MIT Press, 2nd edition, 2000.
- Tank, A., Covert, I., Foti, N., Shojaie, A., and Fox, E. B. Neural Granger causality. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(8):4267–4279, 2022.
- von Büna, P., Meinecke, F. C., Király, F. C., and Müller, K.-R. Finding stationary subspaces in multivariate time series. *Physical Review Letters*, 103(21):214101, 2009.
- Wang, H., Yan, S., Xu, D., Tang, X., and Huang, T. Trace ratio vs. ratio trace for dimensionality reduction. In *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–8, 2007.
- Wiskott, L. and Sejnowski, T. J. Slow feature analysis: Unsupervised learning of invariances. *Neural Computation*, 14(4):715–770, 2002.

A. Detailed Proofs of Theoretical Results

A.1. Proof of Lemma 5.3

Properties of $\mathbf{A}_y$.

(i) *Positive semidefiniteness.* For any $\mathbf{w} \in \mathbb{R}^d$,

$$\mathbf{w}^\top \mathbf{A}_y \mathbf{w} = \mathbf{w}^\top \left(\sum_{i=1}^L \Sigma_{zy_i} \sigma_{y_i}^{-2} \Sigma_{y_i z} \right) \mathbf{w} = \sum_{i=1}^L \frac{(\mathbf{w}^\top \Sigma_{zy_i})^2}{\sigma_{y_i}^2} \geq 0.$$

The last equality uses $\Sigma_{y_i z} = \Sigma_{zy_i}^\top$ and the fact that $\Sigma_{zy_i} \in \mathbb{R}^d$ is a vector, so $\mathbf{w}^\top \Sigma_{zy_i}$ is a scalar.

(ii) *Rank bound.* Each term $\Sigma_{zy_i} \sigma_{y_i}^{-2} \Sigma_{y_i z} = \sigma_{y_i}^{-2} \Sigma_{zy_i} \Sigma_{zy_i}^\top$ is a rank-1 matrix (outer product of Σ_{zy_i} with itself). Since $\mathbf{A}_y$ is a sum of L rank-1 matrices, $\text{rank}(\mathbf{A}_y) \leq L$.

(iii) *Trace formula.* We have

$$\text{Tr}(\mathbf{A}_y) = \sum_{i=1}^L \sigma_{y_i}^{-2} \text{Tr}(\Sigma_{zy_i} \Sigma_{y_i z}) = \sum_{i=1}^L \sigma_{y_i}^{-2} \|\Sigma_{zy_i}\|_2^2 = \|\Sigma_{zy} \mathbf{D}_y^{-1/2}\|_F^2,$$

where $\mathbf{D}_y = \text{diag}(\sigma_{y_1}^2, \dots, \sigma_{y_L}^2)$ and $\Sigma_{zy} = [\Sigma_{zy_1}, \dots, \Sigma_{zy_L}] \in \mathbb{R}^{d \times L}$.

If $\text{Cov}(\mathbf{z}) = \mathbf{I}$, then the coefficient of determination for regressing y_i on $\mathbf{z}$ is

$$R^2(y_i | \mathbf{z}) = \frac{\Sigma_{y_i z} \text{Cov}(\mathbf{z})^{-1} \Sigma_{zy_i}}{\sigma_{y_i}^2} = \frac{\|\Sigma_{zy_i}\|_2^2}{\sigma_{y_i}^2},$$

so $\text{Tr}(\mathbf{A}_y) = \sum_{i=1}^L R^2(y_i | \mathbf{z})$.

Properties of $\mathbf{A}_e$.

(i) *Positive semidefiniteness.* The matrix $\mathbf{A}_e = \sum_{k=1}^K \pi_k (\boldsymbol{\mu}_z^{(k)} - \bar{\boldsymbol{\mu}}_z)(\boldsymbol{\mu}_z^{(k)} - \bar{\boldsymbol{\mu}}_z)^\top$ is a weighted sum of outer products, hence positive semidefinite.

(ii) *Rank bound.* Since $\sum_{k=1}^K \pi_k (\boldsymbol{\mu}_z^{(k)} - \bar{\boldsymbol{\mu}}_z) = \bar{\boldsymbol{\mu}}_z - \bar{\boldsymbol{\mu}}_z = \mathbf{0}$, the K centered mean vectors satisfy one linear constraint and thus span at most a $(K-1)$ -dimensional subspace. Hence $\text{rank}(\mathbf{A}_e) \leq K-1$.

(iii) *Law of total variance.* Let e be a random environment with $\mathbb{P}(e = e_k) = \pi_k$. Then $\mathbf{A}_e = \text{Var}(\mathbb{E}[\mathbf{z} | e])$, and the law of total variance gives

$$\text{Var}(\mathbf{z}) = \mathbb{E}[\text{Var}(\mathbf{z} | e)] + \text{Var}(\mathbb{E}[\mathbf{z} | e]) = \mathbb{E}[\text{Var}(\mathbf{z} | e)] + \mathbf{A}_e.$$

(iv) *Mean-invariant directions.* If $\mathbf{w}^\top \boldsymbol{\mu}_z^{(k)} = c$ (constant) for all k , then $\mathbf{w}^\top \bar{\boldsymbol{\mu}}_z = \sum_k \pi_k \mathbf{w}^\top \boldsymbol{\mu}_z^{(k)} = c$. Thus $\mathbf{w}^\top (\boldsymbol{\mu}_z^{(k)} - \bar{\boldsymbol{\mu}}_z) = 0$ for all k , and

$$\mathbf{w}^\top \mathbf{A}_e \mathbf{w} = \sum_{k=1}^K \pi_k (\mathbf{w}^\top (\boldsymbol{\mu}_z^{(k)} - \bar{\boldsymbol{\mu}}_z))^2 = 0.$$

A.2. Proof of Theorem 5.4

Define the generalized Rayleigh quotient

$$R(\mathbf{w}) = \frac{\mathbf{w}^\top \mathbf{A}_y \mathbf{w}}{\mathbf{w}^\top (\mathbf{A}_e + \epsilon \mathbf{I}) \mathbf{w}}.$$

Step 1: Lower bound on $\mathcal{C}$. For any $\mathbf{w} \in \mathcal{C} \setminus \{0\}$, Assumption 5.1 states that $\mathcal{C} \subseteq \ker(\mathbf{A}_e)$, which directly implies $\mathbf{w}^\top \mathbf{A}_e \mathbf{w} = 0$. By the predictive richness condition, $\mathbf{w}^\top \mathbf{A}_y \mathbf{w} \geq \gamma_c \|\mathbf{w}\|^2$. Therefore,

$$R(\mathbf{w}) = \frac{\mathbf{w}^\top \mathbf{A}_y \mathbf{w}}{\mathbf{w}^\top (\mathbf{A}_e + \epsilon \mathbf{I}) \mathbf{w}} = \frac{\mathbf{w}^\top \mathbf{A}_y \mathbf{w}}{\epsilon \|\mathbf{w}\|^2} \geq \frac{\gamma_c}{\epsilon} \rightarrow \infty \quad (\epsilon \rightarrow 0).$$

Step 2: Upper bound on $\mathcal{S} \oplus \mathcal{R}$. Let $\mathbf{w} = \mathbf{w}_s + \mathbf{w}_r$ with $\mathbf{w}_s \in \mathcal{S}$, $\mathbf{w}_r \in \mathcal{R}$, not both zero. Since $\mathcal{R} \subseteq \ker(\mathbf{A}_y) \cap \ker(\mathbf{A}_e)$ (Assumption 5.1(iii)), we have $\mathbf{A}_y \mathbf{w}_r = \mathbf{A}_e \mathbf{w}_r = \mathbf{0}$, so all cross terms vanish and

$$R(\mathbf{w}) = \frac{\mathbf{w}_s^\top \mathbf{A}_y \mathbf{w}_s}{\mathbf{w}_s^\top \mathbf{A}_e \mathbf{w}_s + \epsilon \|\mathbf{w}\|^2} \leq \frac{\|\mathbf{A}_y\|_2 \|\mathbf{w}_s\|^2}{\sigma_s \|\mathbf{w}_s\|^2} = \frac{\|\mathbf{A}_y\|_2}{\sigma_s}$$

whenever $\mathbf{w}_s \neq \mathbf{0}$ (using the environment diversity condition on $\mathcal{S}$), and $R(\mathbf{w}) = 0$ when $\mathbf{w}_s = \mathbf{0}$. In either case R is bounded on $\mathcal{S} \oplus \mathcal{R}$ uniformly in ϵ ; on $\mathcal{R}$ itself the quotient is exactly zero, which accounts for the zero eigenvalues in the statement of the theorem.

Step 3: Spectral separation via min–max. By the Courant–Fischer min–max theorem for generalized eigenvalues,

$$\lambda_j = \max_{\dim(V)=j} \min_{\mathbf{w} \in V \setminus \{0\}} R(\mathbf{w}).$$

For $j \leq r_c = \dim(\mathcal{C})$, we can choose a j -dimensional subspace $V \subseteq \mathcal{C}$. For any $\mathbf{w} \in V \setminus \{0\}$, Step 1 gives $R(\mathbf{w}) \geq \gamma_c/\epsilon$. Thus $\lambda_j \geq \gamma_c/\epsilon$ for $j \leq r_c$.

For $j = r_c + 1$, any $(r_c + 1)$ -dimensional subspace V must intersect the complement $\mathcal{S} \oplus \mathcal{R}$ nontrivially (since $\dim(\mathcal{C}) = r_c$). Let $\mathbf{w} \in V \cap (\mathcal{S} \oplus \mathcal{R}) \setminus \{0\}$. By Step 2, $R(\mathbf{w}) \leq \|\mathbf{A}_y\|_2/\sigma_s$. Thus $\lambda_{r_c+1} \leq \|\mathbf{A}_y\|_2/\sigma_s$.

As $\epsilon \rightarrow 0$, the top r_c eigenvalues diverge like $1/\epsilon$ while λ_{r_c+1} remains bounded. For the subspace-convergence claim, a Rayleigh-quotient argument is *not* sufficient: a direction $\mathbf{w} = (\mathbf{w}_c + \mathbf{w}_r)/\sqrt{2}$ with $\mathbf{w}_r \in \mathcal{R}$ and $\|\mathbf{w}_r\| = \|\mathbf{w}_c\| = 1$ still satisfies $R(\mathbf{w}) \geq \gamma_c/(2\epsilon)$, because the $\mathcal{R}$ -component contributes nothing to the numerator and only $\epsilon \|\mathbf{w}_r\|^2$ to the denominator; eigenvalue divergence alone therefore cannot rule out large $\mathcal{R}$ -components. We argue instead from the eigenvalue equation, in three steps.

(i) *Eigenvectors with $\lambda \neq 0$ lie exactly in $\mathcal{R}^\perp$.* From $\mathbf{A}_y \mathbf{w} = \lambda(\mathbf{A}_e + \epsilon \mathbf{I}) \mathbf{w}$ we obtain $\lambda \epsilon \mathbf{w} = \mathbf{A}_y \mathbf{w} - \lambda \mathbf{A}_e \mathbf{w} \in \text{range}(\mathbf{A}_y) + \text{range}(\mathbf{A}_e)$. Since $\mathbf{A}_y$ and $\mathbf{A}_e$ are symmetric and $\mathcal{R} \subseteq \ker(\mathbf{A}_y) \cap \ker(\mathbf{A}_e)$, we have $\text{range}(\mathbf{A}_y) = \ker(\mathbf{A}_y)^\perp \subseteq \mathcal{R}^\perp$ and likewise $\text{range}(\mathbf{A}_e) \subseteq \mathcal{R}^\perp$. Hence $\mathbf{w} \in \mathcal{R}^\perp$ exactly, for every $\epsilon > 0$.

(ii) *The $\mathcal{S}$ -component vanishes at rate $O(\sqrt{\epsilon})$.* Write $\mathbf{w} = \mathbf{w}_c + \mathbf{w}_s + \mathbf{w}_r$ along $\mathcal{C} \oplus \mathcal{S} \oplus \mathcal{R}$ with $\|\mathbf{w}\| = 1$. Because $\mathcal{C}, \mathcal{R} \subseteq \ker(\mathbf{A}_e)$ and $\mathbf{A}_e$ is symmetric positive semidefinite, $\mathbf{w}^\top \mathbf{A}_e \mathbf{w} = \mathbf{w}_s^\top \mathbf{A}_e \mathbf{w}_s \geq \sigma_s \|\mathbf{w}_s\|^2$. For a top eigenvector, $\lambda \geq \gamma_c/\epsilon$ and $\lambda(\mathbf{w}^\top \mathbf{A}_e \mathbf{w} + \epsilon) = \mathbf{w}^\top \mathbf{A}_y \mathbf{w} \leq \|\mathbf{A}_y\|_2$, so

$$\sigma_s \|\mathbf{w}_s\|^2 \leq \mathbf{w}^\top \mathbf{A}_e \mathbf{w} \leq \frac{\|\mathbf{A}_y\|_2}{\lambda} \leq \frac{\epsilon \|\mathbf{A}_y\|_2}{\gamma_c}, \quad \text{i.e.} \quad \|\mathbf{w}_s\| = O(\sqrt{\epsilon}).$$

(iii) *Convergence under $\mathcal{C} \perp \mathcal{R}$.* By (i), $0 = \langle \mathbf{w}, \mathbf{w}_r \rangle = \langle \mathbf{w}_c, \mathbf{w}_r \rangle + \langle \mathbf{w}_s, \mathbf{w}_r \rangle + \|\mathbf{w}_r\|^2$. With $\mathcal{C} \perp \mathcal{R}$ the first term vanishes, so $\|\mathbf{w}_r\|^2 \leq \|\mathbf{w}_s\| \|\mathbf{w}_r\|$, i.e. $\|\mathbf{w}_r\| \leq \|\mathbf{w}_s\| = O(\sqrt{\epsilon})$. Hence every top eigenvector converges to $\mathcal{C}$, and the top- r_c eigenspace converges to $\mathcal{C}$.

Without $\mathcal{C} \perp \mathcal{R}$, steps (i)–(ii) still hold, so the limit set of top eigenvectors is contained in $(\mathcal{C} \oplus \mathcal{R}) \cap \mathcal{R}^\perp$; its deviation from $\mathcal{C}$ is governed by the principal angles between $\mathcal{C}$ and $\mathcal{R}$ alone—the orientation of $\mathcal{S}$ is irrelevant. In particular, if $\mathcal{C} \perp \mathcal{R}$ the limit is exactly $\mathcal{C}$ even when $\mathcal{S}$ is oblique to both.

A.3. Proof of Corollary 5.5

From Step 1 of Theorem 5.4, $\lambda_{r_c} \geq \gamma_c/\epsilon$.

From Step 2 and the min–max characterization, $\lambda_{r_c+1} \leq \|\mathbf{A}_y\|_2/\sigma_s$.

Taking the ratio:

$$\frac{\lambda_{r_c}}{\lambda_{r_c+1}} \geq \frac{\gamma_c/\epsilon}{\|\mathbf{A}_y\|_2/\sigma_s} = \frac{\gamma_c \sigma_s}{\epsilon \|\mathbf{A}_y\|_2}.$$

A.4. Proof of Proposition 5.8

If $\text{Cov}(X_{j,t-k}, y_{i,t+1}) \neq 0$, then the corresponding entry of Σ_{zy_i} is nonzero. Specifically, since $\mathbf{z}_t = [\mathbf{x}_t^\top, \mathbf{x}_{t-1}^\top, \dots, \mathbf{x}_{t-s}^\top]^\top$, the coordinate $X_{j,t-k}$ corresponds to index $j + km$ in $\mathbf{z}_t$. Thus $[\Sigma_{zy_i}]_{j+km} = \text{Cov}(X_{j,t-k}, y_{i,t+1}) \neq 0$.

Now,

$$e_{j,k}^\top \mathbf{A}_y e_{j,k} = \sum_{\ell=1}^L \frac{[\Sigma_{zy_\ell}]_{j+km}^2}{\sigma_{y_\ell}^2}.$$

The term for $\ell = i$ is $[\Sigma_{zy_i}]_{j+km}^2 / \sigma_{y_i}^2 > 0$. Since all terms are non-negative, the sum is strictly positive.

A.5. Proof of Proposition 5.9

The annihilation condition is the crux of the argument. Write $\mathbf{z} = \mathbf{z}_c + \mathbf{z}_s + \mathbf{z}_r$ per Assumption 5.1. By the hypothesis that $\mathbf{W}_s^\top \mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in \mathcal{S} \oplus \mathcal{R}$ (which holds in particular when the decomposition is orthogonal and $\mathbf{W}_s$ spans $\mathcal{C}$),

$$\mathbf{s} = \mathbf{W}_s^\top \mathbf{z} = \mathbf{W}_s^\top \mathbf{z}_c + \underbrace{\mathbf{W}_s^\top (\mathbf{z}_s + \mathbf{z}_r)}_{=\mathbf{0}} = \mathbf{W}_s^\top \mathbf{z}_c,$$

so $\mathbf{s}$ is a function of $\mathbf{z}_c$ alone. Without this condition, $\mathbf{s}$ retains a component of $\mathbf{z}_s$, whose distribution depends on e , and the conclusion fails in general.

We now argue via the joint distribution, which requires no injectivity of $\mathbf{W}_s$ on $\mathcal{C}$ (a property that the annihilation hypothesis alone does not guarantee; e.g., $\mathbf{W}_s$ could have fewer than $\dim(\mathcal{C})$ columns). Under the structural equation model, the pair

$$(\mathbf{Y}, \mathbf{s}) = (f(\mathbf{z}_c) + \boldsymbol{\eta}, \mathbf{W}_s^\top \mathbf{z}_c)$$

is a measurable function of $(\mathbf{z}_c, \boldsymbol{\eta})$, and the map itself does not depend on e (neither f nor $\mathbf{W}_s$ does). Since $(\mathbf{z}_c, \boldsymbol{\eta}) \perp\!\!\!\perp e$, the joint law of $(\mathbf{z}_c, \boldsymbol{\eta})$ is identical in every environment, and hence so is the joint law of $(\mathbf{Y}, \mathbf{s})$. The conditional law $P(\mathbf{Y} \mid \mathbf{s})$ is a functional of the joint law of $(\mathbf{Y}, \mathbf{s})$; therefore $P^e(\mathbf{Y} \mid \mathbf{s}) = P^{e'}(\mathbf{Y} \mid \mathbf{s})$ for all $e, e' \in \mathcal{E}$. (When $\mathbf{W}_s$ is injective on $\mathcal{C}$ —as under the first hypothesis branch—the conclusion can equivalently be obtained by recovering $\mathbf{z}_c = g(\mathbf{s})$, but injectivity is not needed.)

A.6. Proof of Proposition 5.10

The CFL algorithm consists of the following steps:

Step 1: Time-lagged matrix construction. For each of the $N - s$ valid time indices, we concatenate $s + 1$ vectors of dimension m . This requires $O((N - s) \cdot m \cdot (s + 1)) = O(N \cdot m \cdot s)$ operations.

Step 2: Covariance estimation. Computing $\hat{\mathbf{A}}_y$ requires: (i) sample cross-covariance $\hat{\Sigma}_{zy} \in \mathbb{R}^{d \times L}$: $O(N \cdot d \cdot L)$; (ii) forming $\hat{\mathbf{A}}_y = \sum_{i=1}^L \hat{\Sigma}_{zy_i} \hat{\sigma}_{y_i}^{-2} \hat{\Sigma}_{y_i z}$: $O(L \cdot d^2)$. Computing $\hat{\mathbf{A}}_e$ requires: (i) sample means per environment: $O(N \cdot d)$; (ii) forming outer products: $O(K \cdot d^2)$. Total for covariance estimation: $O(N \cdot d^2)$ (assuming $L, K = O(d)$).

Step 3: Generalized eigenvalue problem. Solving $\mathbf{A}_y \mathbf{w} = \lambda \mathbf{B} \mathbf{w}$ for $d \times d$ matrices using standard algorithms (e.g., QZ algorithm) requires $O(d^3)$ operations.

Total complexity: $O(Nms + Nd^2 + d^3) = O(Nd^2 + d^3)$, since $d = m(s + 1)$ implies $Nms = O(Nd)$.

For $N \gg d$, the $O(Nd^2)$ covariance estimation term dominates.

B. Additional Theoretical Results

B.1. Optimality Conditions for CFL

Proposition B.1 (KKT Conditions). *The generalized eigenvalue problem (15) is equivalent to the KKT conditions of:*

$$\max_{\mathbf{w}} \mathbf{w}^\top \mathbf{A}_y \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top (\mathbf{A}_e + \epsilon \mathbf{I}) \mathbf{w} = 1. \quad (23)$$

Proof. The Lagrangian is $\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^\top \mathbf{A}_y \mathbf{w} - \lambda(\mathbf{w}^\top (\mathbf{A}_e + \epsilon \mathbf{I}) \mathbf{w} - 1)$. Stationarity $\nabla_{\mathbf{w}} \mathcal{L} = 0$ gives $\mathbf{A}_y \mathbf{w} = \lambda(\mathbf{A}_e + \epsilon \mathbf{I}) \mathbf{w}$. $\square$

Table 1. Fault detection performance across the 18 detectable TEP faults under the common validation-target protocol. Rows give per-fault FDR (%); summary rows give average FDR (%), realized FAR (%), and mean detection delay, all computed over the 18 faults shown. Green marks the best rounded FDR for each fault.

Fault	PCA	SFA	DiPLS	DiPCA	DiCCA	Maha	AE	VAE	LSTM-AE	OCSVM	KPCA	LOF	IF	GMM	GEV-SFA	CFL
F01	99.8	99.9	99.5	99.8	99.8	99.8	99.6	99.9	99.4	99.6	99.6	99.6	85.1	99.8	99.6	99.8
F02	98.9	98.8	98.6	98.8	98.8	98.9	98.6	98.6	98.4	98.6	98.6	98.6	97.5	98.9	98.8	98.8
F04	99.8	100.0	66.8	100.0	99.8	100.0	95.0	96.9	100.0	81.0	86.8	95.6	13.5	98.5	100.0	99.8
F05	31.0	100.0	33.5	31.5	100.0	100.0	30.0	30.1	35.9	31.5	33.1	31.0	33.9	100.0	39.4	100.0
F06	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	99.9	100.0	100.0	100.0	99.2	100.0	100.0	100.0
F07	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	47.9	100.0	100.0	100.0
F08	98.0	97.8	98.1	98.2	98.1	98.1	98.0	97.9	98.2	98.4	98.4	98.5	97.6	98.8	98.1	97.9
F10	61.5	89.5	54.1	54.8	61.8	90.4	56.0	54.0	67.1	58.2	59.5	58.4	44.5	89.1	70.1	93.4
F11	75.6	82.8	60.6	72.9	73.9	79.2	71.8	73.6	86.5	66.8	67.8	70.5	21.1	68.9	94.8	78.1
F12	99.2	99.9	99.1	99.0	99.9	99.9	99.0	98.9	100.0	99.2	99.5	99.1	97.9	99.9	99.6	99.9
F13	95.5	95.6	94.9	95.8	95.5	95.6	95.6	95.6	94.9	95.5	95.4	95.5	94.8	95.5	95.8	95.6
F14	100.0	100.0	99.9	100.0	100.0	100.0	100.0	100.0	99.9	100.0	100.0	100.0	11.4	99.9	99.9	99.9
F16	56.8	92.2	38.2	51.0	55.4	93.9	46.6	48.6	62.5	49.9	50.9	50.2	30.8	92.8	56.5	96.8
F17	96.0	97.1	87.1	95.4	95.5	96.2	91.6	92.6	96.6	90.1	90.0	92.0	36.2	94.2	97.4	97.4
F18	90.5	90.5	90.1	90.8	91.4	91.1	90.2	90.8	89.8	89.8	90.1	90.1	89.8	91.6	91.4	90.8
F19	26.0	91.4	5.9	24.2	37.2	90.5	15.1	22.6	16.4	8.1	8.6	13.1	7.2	78.4	50.9	90.0
F20	63.0	90.6	55.4	62.6	71.0	91.0	57.8	57.4	70.2	58.9	59.5	59.0	42.5	89.6	80.8	90.5
F21	52.8	57.8	43.1	51.7	56.5	61.6	44.8	45.9	48.2	42.1	43.2	42.0	41.0	63.4	61.6	74.0
Avg FDR	80.2	93.5	73.6	79.2	85.2	93.7	77.2	78.0	81.3	76.0	76.7	77.4	55.1	92.2	85.2	94.6
Avg FAR	3.2	3.0	3.3	4.7	5.0	2.0	1.8	2.4	2.4	2.0	2.7	1.5	3.8	2.5	3.7	1.3
Delay	34	30	48	31	35	26	38	33	34	41	37	38	96	28	26	24

B.2. Relationship to Correlation/CCA

Proposition B.2 (Correlation Interpretation under Whitening). *When $L = 1$ and $\mathbf{A}_e = \mathbf{0}$, CFL reduces to*

$$\max_{\mathbf{w} \neq \mathbf{0}} \frac{(\mathbf{w}^\top \boldsymbol{\Sigma}_{zy})^2}{\mathbf{w}^\top \mathbf{w} \cdot \sigma_y^2}. \quad (24)$$

If additionally $\text{Cov}(\mathbf{z}) = \mathbf{I}$ (whitened $\mathbf{z}$), this is equivalent to maximizing $\text{Corr}^2(\mathbf{w}^\top \mathbf{z}, y)$. Without whitening, the exact correlation objective uses $\mathbf{w}^\top \text{Cov}(\mathbf{z}) \mathbf{w}$ in the denominator.

B.3. Extension to Nonlinear Features

Remark B.3 (Kernel CFL). The CFL framework can be extended to nonlinear features via the kernel trick. Define feature map $\phi : \mathbb{R}^d \rightarrow \mathcal{H}$ and kernel $k(\mathbf{z}, \mathbf{z}') = \langle \phi(\mathbf{z}), \phi(\mathbf{z}') \rangle_{\mathcal{H}}$. Kernel CFL can penalize environment mean shifts in RKHS by matching mean embeddings, yielding kernelized analogues of $\mathbf{A}_y$ and $\mathbf{A}_e$.

C. Experimental Details

Figure 4 shows the corresponding CFL monitoring plots. Figures 5–9 present the monitoring plots for the five classical baselines (PCA, SFA, DiPLS, DiPCA, and DiCCA) on the 18 detectable faults, and Figure 10 summarizes the average FDR comparison over the same faults. One important difference from the corresponding figures in earlier versions of this work: the red dashed control limits shown here are the *calibrated* thresholds obtained on the independent normal validation run under the same joint 5% OR-alarm target used throughout the paper. Nominal parametric (F/χ^2) limits are severely miscalibrated on this data (for several methods, nominal 95% limits yield test false-alarm rates above 60%) and would render such plots visually misleading; all quantitative comparisons in this paper likewise use the calibrated thresholds.

Table 2. Ablations of CFL components and hyper-parameters under the calibrated protocol. Each cell reports average FDR / realized FAR (%) over the 18 detectable faults.

LAG ORDER s	0	2	5	8
FDR / FAR	92.5 / 2.1	93.3 / 1.5	94.6 / 1.3	95.0 / 1.2
STABLE DIM. d_s	4	8	12	
FDR / FAR	94.5 / 1.7	94.5 / 1.6	94.6 / 1.3	
VARYING DIM. d_v	0	5	9	
FDR / FAR	92.1 / 1.7	92.9 / 1.1	94.6 / 1.3	
ENVIRONMENTS K	3	5	10	20
FDR / FAR	94.8 / 3.2	94.2 / 1.8	94.6 / 1.3	94.0 / 2.1
REGULARIZ. ϵ	10^{-5}	10^{-4}	10^{-3}	10^{-2}
FDR / FAR	94.5 / 1.4	94.6 / 1.3	94.3 / 1.2	93.8 / 1.4
COMPONENTS	FULL	NO $\mathbf{A}_e$	NO $\mathbf{A}_y$	NO T_{sf}^2
FDR / FAR	94.6 / 1.3	93.6 / 2.6	94.6 / 2.6	91.7 / 1.7

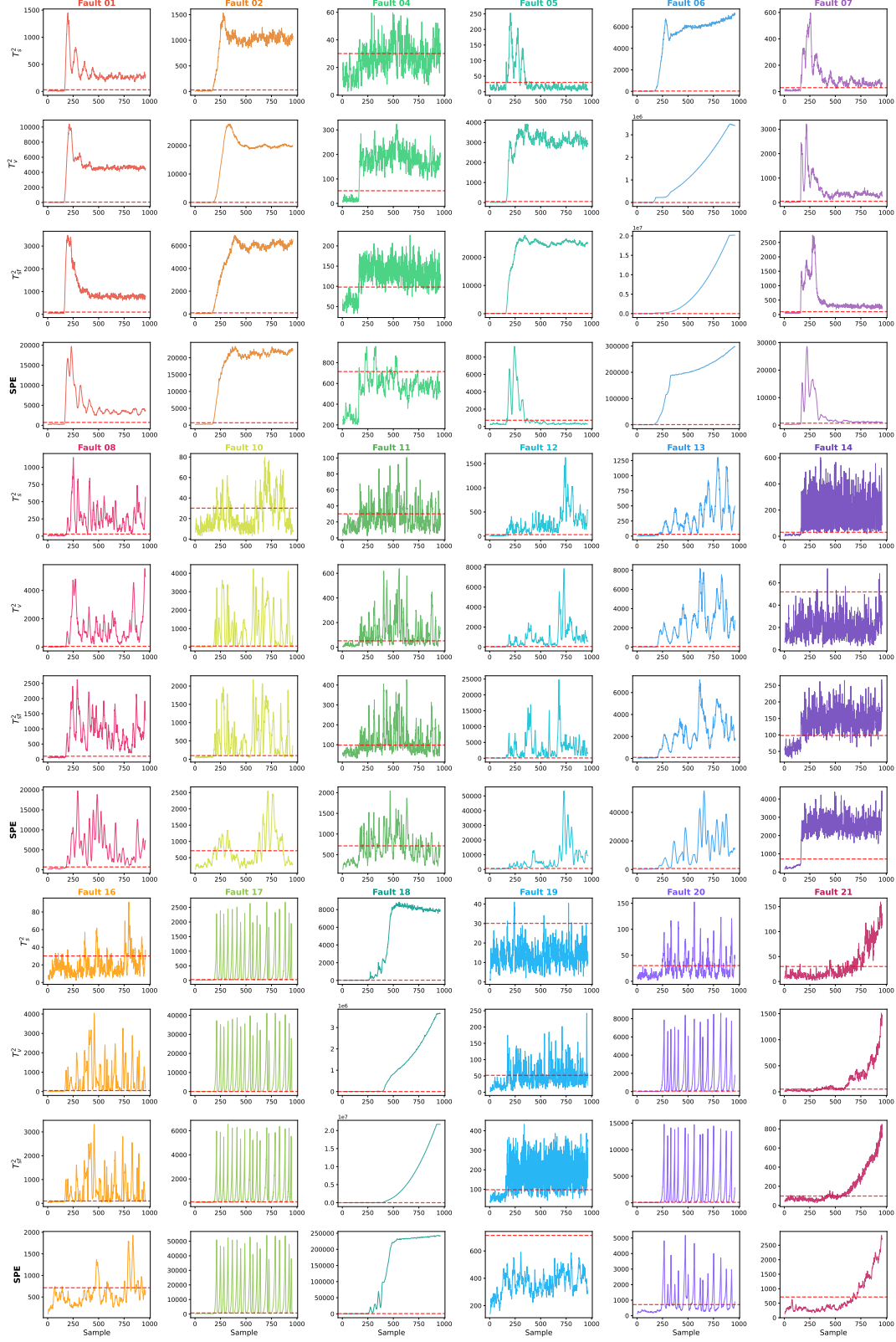


Figure 4. CFL monitoring results for the 18 detectable TEP faults. For each fault (columns), the four monitoring statistics T_s^2 , T_v^2 , T_{sf}^2 , and SPE are shown in separate rows; red dashed lines are the calibrated thresholds obtained on the normal validation run, and the fault is introduced at sample 160.

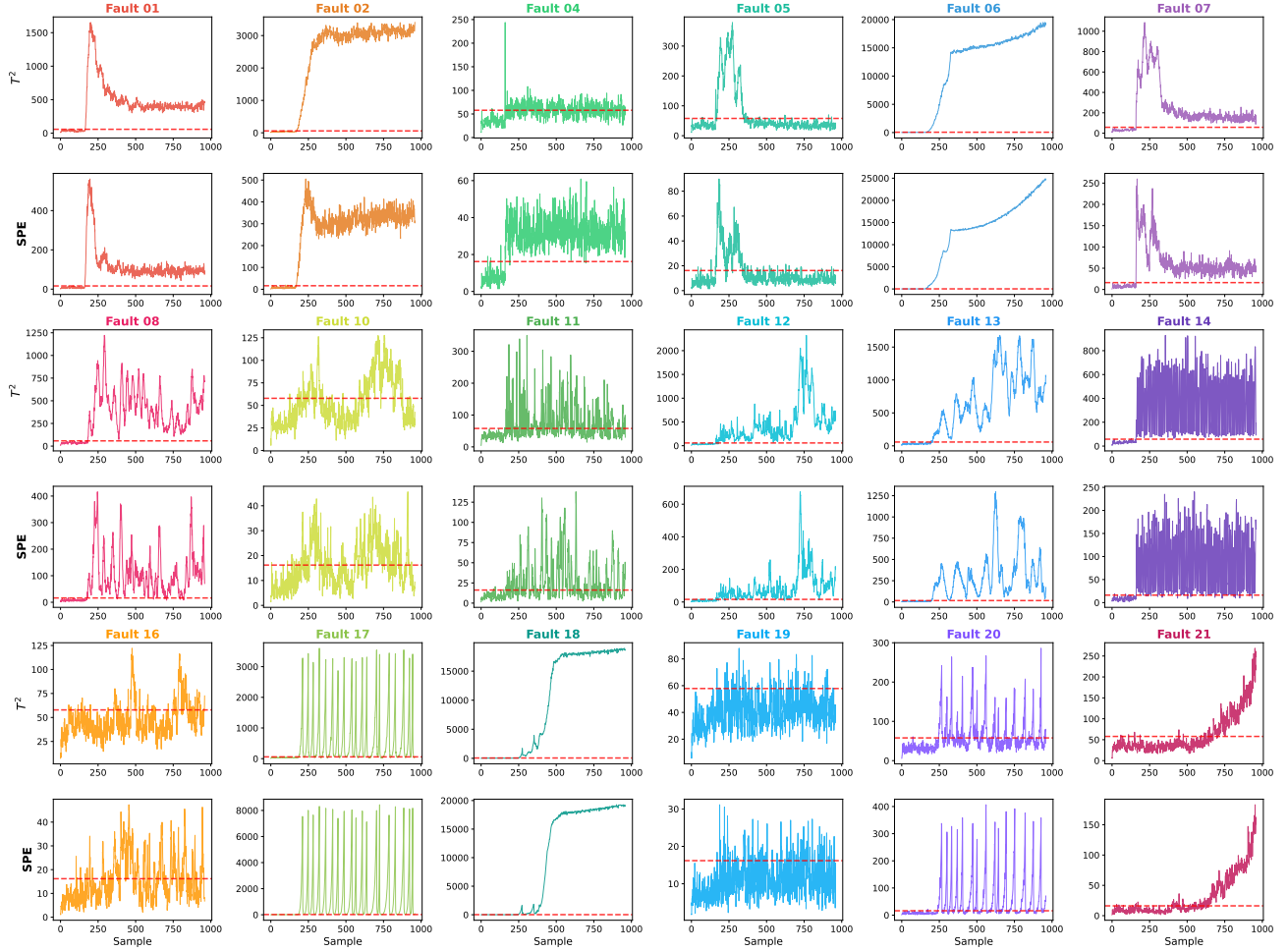


Figure 5. PCA fault detection monitoring results for the 18 detectable TEP faults (T^2 and SPE). Red dashed lines are thresholds calibrated on the normal validation run (joint 5% OR-alarm target); the fault is introduced at sample 160.

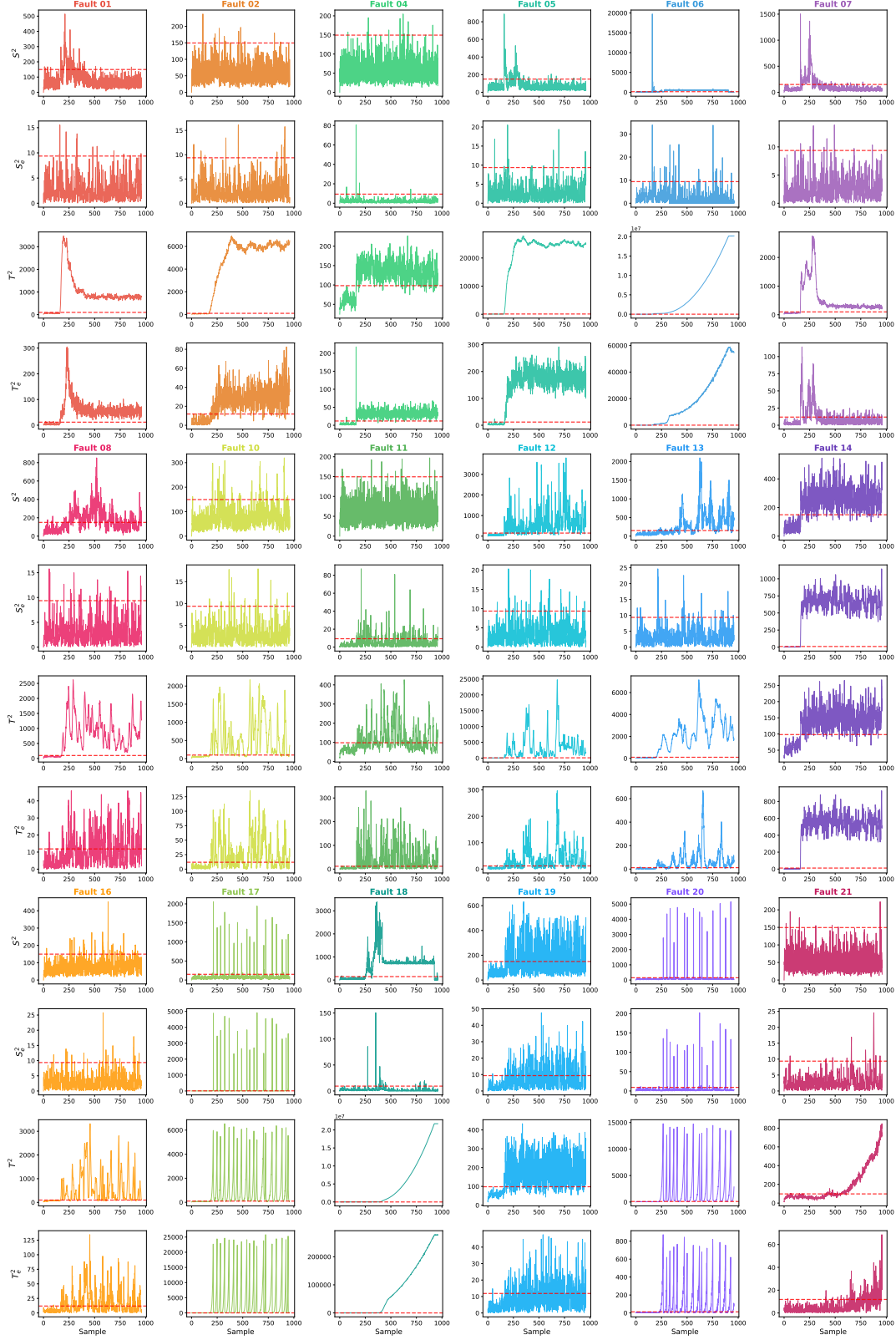


Figure 6. SFA fault detection monitoring results for the 18 detectable TEP faults (S^2 , S_e^2 , T^2 , T_e^2). Red dashed lines are calibrated thresholds; the fault is introduced at sample 160.

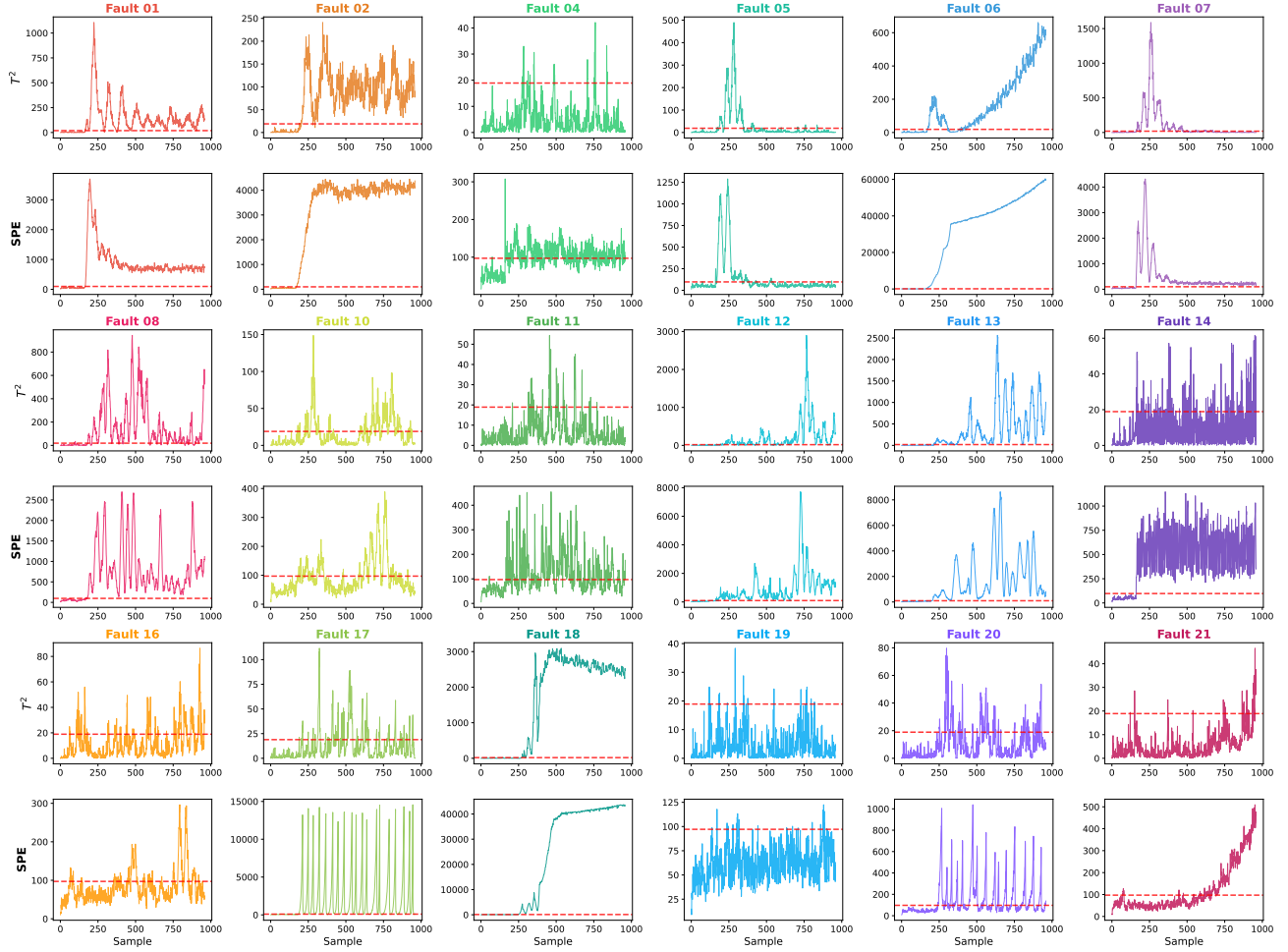


Figure 7. DiPLS fault detection monitoring results for the 18 detectable TEP faults (T^2 and SPE). Red dashed lines are calibrated thresholds; the fault is introduced at sample 160.

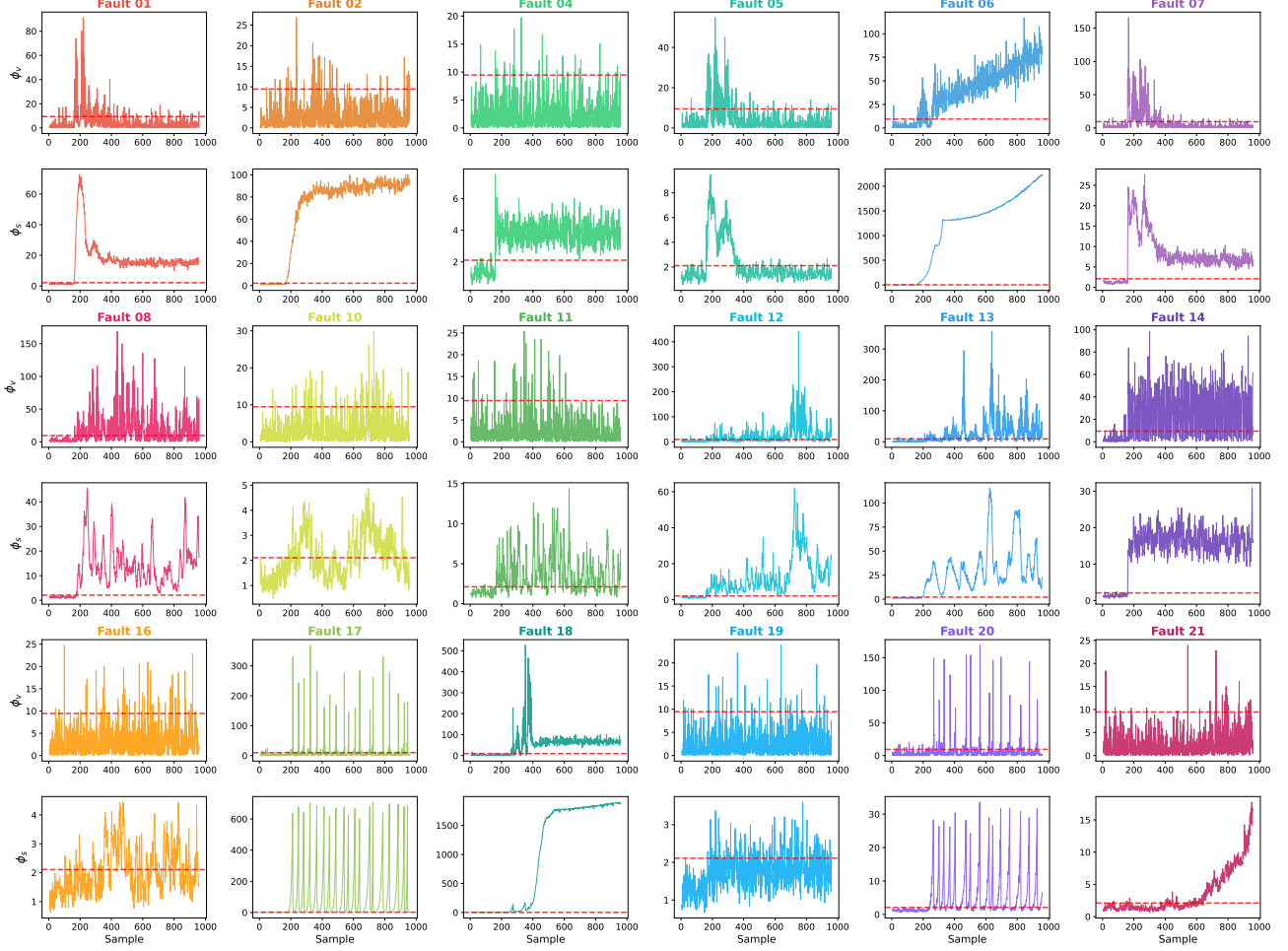


Figure 8. DiPCA fault detection monitoring results for the 18 detectable TEP faults (ϕ_v and ϕ_s). Red dashed lines are calibrated thresholds; the fault is introduced at sample 160.

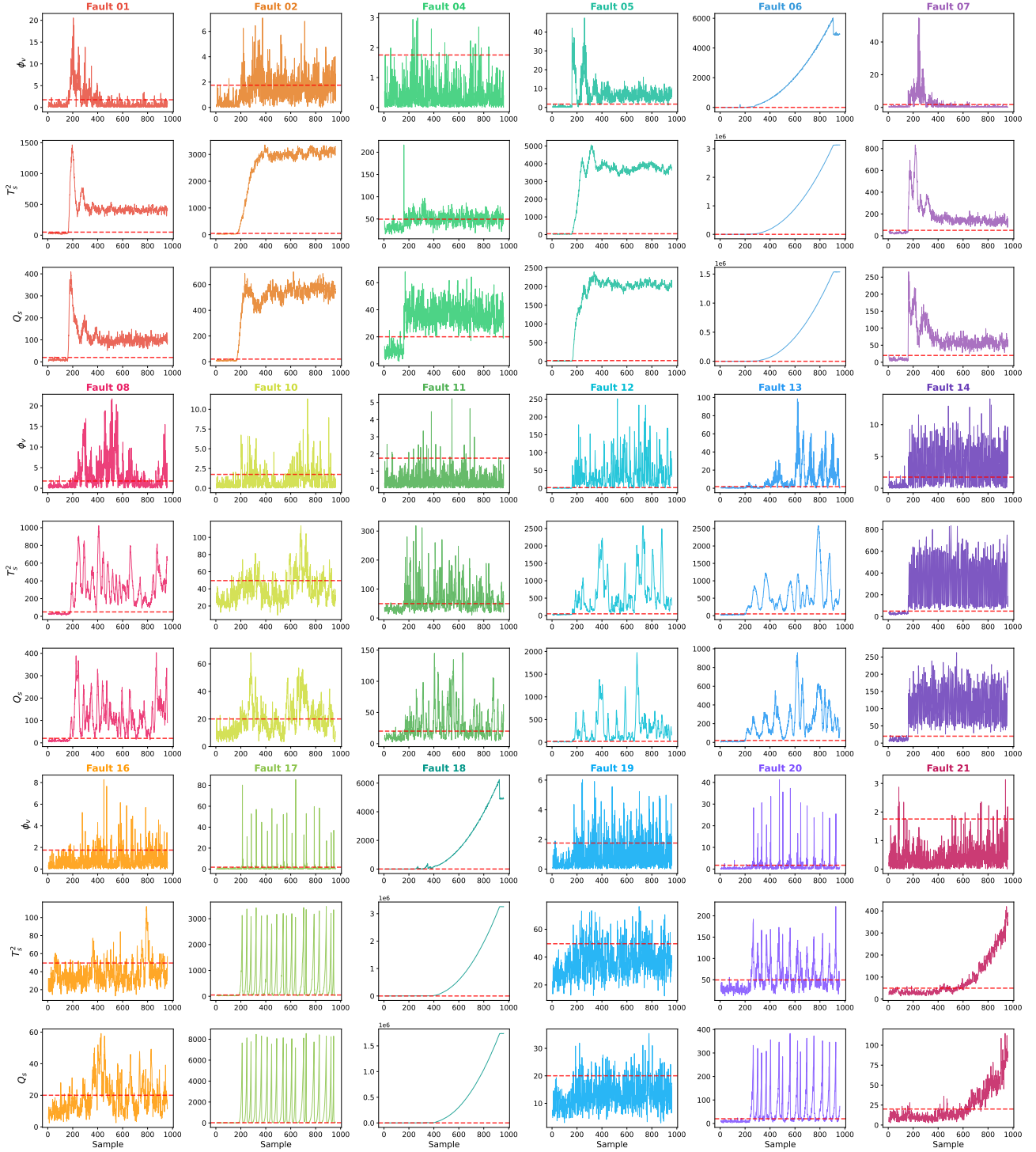


Figure 9. DiCCA fault detection monitoring results for the 18 detectable TEP faults (ϕ_v , T_s^2 , and Q_s^2). Red dashed lines are calibrated thresholds; the fault is introduced at sample 160.

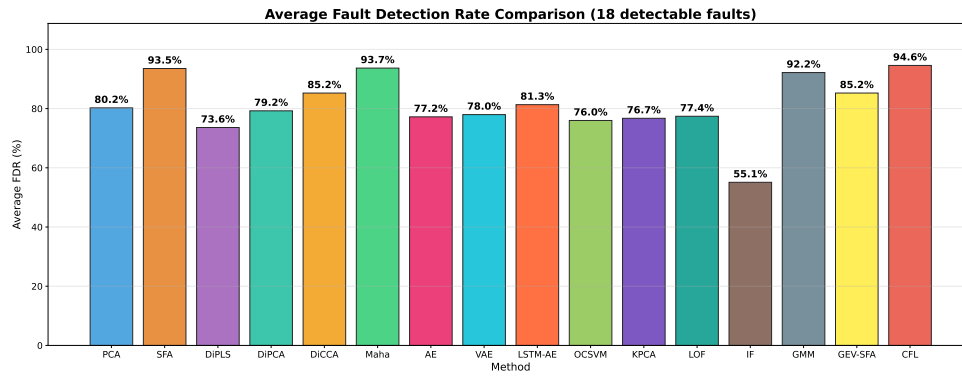


Figure 10. Average FDR comparison across all 16 methods over the 18 detectable TEP faults, under the common calibrated protocol (5% validation OR-alarm target).